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AI, ML, DL

Batch 1

Importing All Necessary libraries

In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import sklearn as sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

Analyzing The Data sets

In [2]:
```
fd_Data = pd.read_csv("./FraudData.csv")
print(fd_Data.head())
print(fd_Data.info())
```

```
   step      type    amount     nameOrig  oldbalanceOrg  newbalanceOrig  \
0     1   PAYMENT   9839.64  C1231006815       170136.0       160296.36
1     1   PAYMENT   1864.28  C1666544295        21249.0        19384.72
2     1  TRANSFER    181.00  C1305486145          181.0            0.00
3     1  CASH_OUT    181.00   C840083671          181.0            0.00
4     1   PAYMENT  11668.14  C2048537720        41554.0        29885.86

      nameDest  oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
0  M1979787155             0.0             0.0        0               0
1  M2044282225             0.0             0.0        0               0
2   C553264065             0.0             0.0        1               0
3    C38997010         21182.0             0.0        1               0
4  M1230701703             0.0             0.0        0               0
<class 'pandas.DataFrame'>
RangeIndex: 6362620 entries, 0 to 6362619
Data columns (total 11 columns):
 #   Column          Dtype
---  ------          -----
 0   step            int64
 1   type            str
 2   amount          float64
 3   nameOrig        str
 4   oldbalanceOrg   float64
 5   newbalanceOrig  float64
 6   nameDest        str
 7   oldbalanceDest  float64
 8   newbalanceDest  float64
 9   isFraud         int64
 10  isFlaggedFraud  int64
dtypes: float64(5), int64(3), str(3)
memory usage: 534.0 MB
None
```

In [4]:

```
fin_df = pd.read_csv("./yFinData.csv")
print(fin_df.head())
print(fin_df.info())
```

```
        Date      Open      High       Low     Close  Adj Close     Volume
0  4/28/2023  33797.43  34104.56  33728.40  34098.16   34098.16  354310000
1  4/29/2023  33381.66  33859.75  33374.65  33826.16   33826.16  343240000
2  4/30/2023  33596.34  33645.83  33235.85  33301.87   33301.87  321170000
3   5/1/2023  33828.34  33875.49  33525.39  33530.83   33530.83  297880000
4   5/2/2023  33805.04  33891.15  33726.09  33875.40   33875.40  252020000
<class 'pandas.DataFrame'>
RangeIndex: 1258 entries, 0 to 1257
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Date       1258 non-null   str
 1   Open       1258 non-null   float64
 2   High       1258 non-null   float64
 3   Low        1258 non-null   float64
 4   Close      1258 non-null   float64
 5   Adj Close  1258 non-null   float64
 6   Volume     1258 non-null   int64
dtypes: float64(5), int64(1), str(1)
memory usage: 68.9 KB
None
```

In [5]:

```
st_df = pd.read_csv("./StPerformData.csv")
print(st_df.head())
print(st_df.info())
```

```
   student_id  age  gender school_type parent_education  study_hours  \
0           1   14    male      public    post graduate          3.1
1           2   18  female      public         graduate          3.7
2           3   17  female     private    post graduate          7.9
3           4   16   other      public      high school          1.1
4           5   16  female      public      high school          1.3

   attendance_percentage internet_access travel_time extra_activities  \
0                   84.3             yes     <15 min              yes
1                   87.8             yes     >60 min               no
2                   65.5              no     <15 min               no
3                   58.1              no   15-30 min               no
4                   61.0             yes   30-60 min              yes

  study_method  math_score  science_score  english_score  overall_score  \
0        notes        42.7           55.4           57.0           53.1
1     textbook        57.6           68.8           64.8           61.3
2        notes        84.8           95.0           79.2           89.6
3        notes        44.4           27.5           54.7           41.6
4  group study         8.9           32.7           30.0           25.4

  final_grade
0           e
1           d
2           b
3           e
4           f
<class 'pandas.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 16 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   student_id             25000 non-null  int64
 1   age                    25000 non-null  int64
 2   gender                 25000 non-null  str
 3   school_type            25000 non-null  str
 4   parent_education       25000 non-null  str
 5   study_hours            25000 non-null  float64
 6   attendance_percentage  25000 non-null  float64
 7   internet_access        25000 non-null  str
 8   travel_time            25000 non-null  str
 9   extra_activities       25000 non-null  str
 10  study_method           25000 non-null  str
 11  math_score             25000 non-null  float64
 12  science_score          25000 non-null  float64
 13  english_score          25000 non-null  float64
 14  overall_score          25000 non-null  float64
 15  final_grade            25000 non-null  str
dtypes: float64(6), int64(2), str(8)
memory usage: 3.1 MB
None
```

Section A (Data Cleaning & Preprocessing)

Question 4: Perform Cleaning & Preprocessng on Sales Dataset

In [6]:
```
# ==============================================================================
# STEP 1: LOAD AND EXPLORE THE DATASET
# ==============================================================================
# Load the sales dataset (using FraudData.csv for this exercise)
sales_df = pd.read_csv("./FraudData.csv")

# Display basic information about the dataset
print("=" * 70)
print("STEP 1: DATASET OVERVIEW")
print("=" * 70)
print(f"\nShape: {sales_df.shape[0]} rows × {sales_df.shape[1]} columns")
print(f"\nColumn Names and Data Types:")
print(sales_df.dtypes)
print(f"\nFirst 5 rows:")
print(sales_df.head())
print(f"\nBasic Statistics:")
print(sales_df.describe())

# Check for missing values
print("\n" + "=" * 70)
print("Missing Values Check:")
print("=" * 70)
missing_values = sales_df.isnull().sum()
print(missing_values[missing_values > 0] if missing_values.sum() > 0 else "No

# Check for duplicate rows
print(f"\nDuplicate Rows: {sales_df.duplicated().sum()}")
print(f"\nDataset Info:")
sales_df.info()
```

```
======================================================================
STEP 1: DATASET OVERVIEW
======================================================================

Shape: 6362620 rows × 11 columns

Column Names and Data Types:
step                int64
type                  str
amount            float64
nameOrig              str
oldbalanceOrg     float64
newbalanceOrig    float64
nameDest              str
oldbalanceDest    float64
newbalanceDest    float64
isFraud             int64
isFlaggedFraud      int64
dtype: object

First 5 rows:
   step      type    amount     nameOrig  oldbalanceOrg  newbalanceOrig  \
0     1   PAYMENT   9839.64  C1231006815       170136.0       160296.36
1     1   PAYMENT   1864.28  C1666544295        21249.0        19384.72
2     1  TRANSFER    181.00  C1305486145          181.0            0.00
3     1  CASH_OUT    181.00   C840083671          181.0            0.00
4     1   PAYMENT  11668.14  C2048537720        41554.0        29885.86

      nameDest  oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
0  M1979787155             0.0             0.0        0               0
1  M2044282225             0.0             0.0        0               0
2   C553264065             0.0             0.0        1               0
3    C38997010         21182.0             0.0        1               0
4  M1230701703             0.0             0.0        0               0

Basic Statistics:
               step        amount  oldbalanceOrg  newbalanceOrig  \
count  6.362620e+06  6.362620e+06   6.362620e+06    6.362620e+06
mean   2.433972e+02  1.798619e+05   8.338831e+05    8.551137e+05
std    1.423320e+02  6.038582e+05   2.888243e+06    2.924049e+06
min    1.000000e+00  0.000000e+00   0.000000e+00    0.000000e+00
25%    1.560000e+02  1.338957e+04   0.000000e+00    0.000000e+00
50%    2.390000e+02  7.487194e+04   1.420800e+04    0.000000e+00
75%    3.350000e+02  2.087215e+05   1.073152e+05    1.442584e+05
max    7.430000e+02  9.244552e+07   5.958504e+07    4.958504e+07

       oldbalanceDest  newbalanceDest       isFraud  isFlaggedFraud
count    6.362620e+06    6.362620e+06  6.362620e+06    6.362620e+06
mean     1.100702e+06    1.224996e+06  1.290820e-03    2.514687e-06
std      3.399180e+06    3.674129e+06  3.590480e-02    1.585775e-03
min      0.000000e+00    0.000000e+00  0.000000e+00    0.000000e+00
25%      0.000000e+00    0.000000e+00  0.000000e+00    0.000000e+00
50%      1.327057e+05    2.146614e+05  0.000000e+00    0.000000e+00
75%      9.430367e+05    1.111909e+06  0.000000e+00    0.000000e+00
```

```
max      3.560159e+08    3.561793e+08  1.000000e+00    1.000000e+00

======================================================================
Missing Values Check:
======================================================================
No missing values found!

Duplicate Rows: 0

Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 6362620 entries, 0 to 6362619
Data columns (total 11 columns):
 #   Column          Dtype
---  ------          -----
 0   step            int64
 1   type            str
 2   amount          float64
 3   nameOrig        str
 4   oldbalanceOrg   float64
 5   newbalanceOrig  float64
 6   nameDest        str
 7   oldbalanceDest  float64
 8   newbalanceDest  float64
 9   isFraud         int64
 10  isFlaggedFraud  int64
dtypes: float64(5), int64(3), str(3)
memory usage: 534.0 MB
```

In [7]:

```python
# ============================================================================
# STEP 2: HANDLE MISSING VALUES
# ============================================================================
# Check for missing values in detail
print("\n" + "=" * 70)
print("STEP 2: HANDLING MISSING VALUES")
print("=" * 70)

missing_percent = (sales_df.isnull().sum() / len(sales_df)) * 100
missing_df = pd.DataFrame({
    'Missing Count': sales_df.isnull().sum(),
    'Missing Percentage': missing_percent
})
print("\nMissing Values Summary:")
print(missing_df[missing_df['Missing Count'] > 0])

# Handle missing values - Fill numerical columns with mean, categorical with m
print("\nFilling missing values...")
for col in sales_df.columns:
    if sales_df[col].isnull().sum() > 0:
        if sales_df[col].dtype in ['float64', 'int64']:
            # Fill numerical columns with median (more robust than mean for ou
            sales_df[col].fillna(sales_df[col].median(), inplace=True)
            print(f"  - {col}: filled with median")
        else:
```

```
            # Fill categorical columns with mode
            sales_df[col].fillna(sales_df[col].mode()[0], inplace=True)
            print(f"  - {col}: filled with mode")

print(f"\nMissing values after cleaning: {sales_df.isnull().sum().sum()}")
```

```
======================================================================
STEP 2: HANDLING MISSING VALUES
======================================================================

Missing Values Summary:
Empty DataFrame
Columns: [Missing Count, Missing Percentage]
Index: []

Filling missing values...

Missing values after cleaning: 0
```

In [8]:

```
# ==============================================================================
# STEP 3: REMOVE DUPLICATES
# ==============================================================================
print("\n" + "=" * 70)
print("STEP 3: REMOVING DUPLICATES")
print("=" * 70)

duplicates_before = sales_df.duplicated().sum()
print(f"\nDuplicate rows found: {duplicates_before}")

if duplicates_before > 0:
    # Remove completely duplicate rows (all columns same)
    sales_df.drop_duplicates(inplace=True)
    duplicates_after = sales_df.duplicated().sum()
    print(f"Duplicate rows after removal: {duplicates_after}")
    print(f"Shape after removing duplicates: {sales_df.shape}")
else:
    print("No duplicate rows found - dataset is clean!")

print(f"Dataset shape: {sales_df.shape[0]} rows × {sales_df.shape[1]} columns"
```

```
======================================================================
STEP 3: REMOVING DUPLICATES
======================================================================

Duplicate rows found: 0
No duplicate rows found - dataset is clean!
Dataset shape: 6362620 rows × 11 columns
```

In [9]:

```
# ==============================================================================
# STEP 4: DETECT AND HANDLE OUTLIERS (IQR Method)
# ==============================================================================
print("\n" + "=" * 70)
print("STEP 4: DETECTING AND HANDLING OUTLIERS")
print("=" * 70)
```

```python
# Select only numerical columns for outlier detection
numerical_cols = sales_df.select_dtypes(include=['float64', 'int64']).columns

print("\nOutlier Detection using IQR (Interquartile Range) method:")
print("-" * 70)

outlier_summary = {}
for col in numerical_cols:
    Q1 = sales_df[col].quantile(0.25)
    Q3 = sales_df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Count outliers
    outliers = ((sales_df[col] < lower_bound) | (sales_df[col] > upper_bound))

    if outliers > 0:
        outlier_summary[col] = outliers
        print(f"\n{col}:")
        print(f"  - Q1: {Q1:.2f}, Q3: {Q3:.2f}, IQR: {IQR:.2f}")
        print(f"  - Lower bound: {lower_bound:.2f}, Upper bound: {upper_bound:
        print(f"  - Outliers found: {outliers}")

        # Cap outliers (replace with bounds instead of removing)
        sales_df[col] = sales_df[col].clip(lower_bound, upper_bound)
        print(f"  - Action: Capped outliers to bounds")

if not outlier_summary:
    print("\nNo significant outliers detected using IQR method.")
```

```
======================================================================
STEP 4: DETECTING AND HANDLING OUTLIERS
======================================================================

Outlier Detection using IQR (Interquartile Range) method:
----------------------------------------------------------------------

step:
  - Q1: 156.00, Q3: 335.00, IQR: 179.00
  - Lower bound: -112.50, Upper bound: 603.50
  - Outliers found: 102688
  - Action: Capped outliers to bounds

amount:
  - Q1: 13389.57, Q3: 208721.48, IQR: 195331.91
  - Lower bound: -279608.29, Upper bound: 501719.34
  - Outliers found: 338078
  - Action: Capped outliers to bounds

oldbalanceOrg:
  - Q1: 0.00, Q3: 107315.18, IQR: 107315.18
  - Lower bound: -160972.76, Upper bound: 268287.94
  - Outliers found: 1112507
  - Action: Capped outliers to bounds

newbalanceOrig:
  - Q1: 0.00, Q3: 144258.41, IQR: 144258.41
  - Lower bound: -216387.61, Upper bound: 360646.03
  - Outliers found: 1053391
  - Action: Capped outliers to bounds

oldbalanceDest:
  - Q1: 0.00, Q3: 943036.71, IQR: 943036.71
  - Lower bound: -1414555.06, Upper bound: 2357591.77
  - Outliers found: 786135
  - Action: Capped outliers to bounds

newbalanceDest:
  - Q1: 0.00, Q3: 1111909.25, IQR: 1111909.25
  - Lower bound: -1667863.88, Upper bound: 2779773.12
  - Outliers found: 738527
  - Action: Capped outliers to bounds

isFraud:
  - Q1: 0.00, Q3: 0.00, IQR: 0.00
  - Lower bound: 0.00, Upper bound: 0.00
  - Outliers found: 8213
  - Action: Capped outliers to bounds

isFlaggedFraud:
  - Q1: 0.00, Q3: 0.00, IQR: 0.00
  - Lower bound: 0.00, Upper bound: 0.00
  - Outliers found: 16
  - Action: Capped outliers to bounds
```

In [10]:

```python
# ==============================================================================
# STEP 5: DATA TYPE OPTIMIZATION
# ==============================================================================
print("\n" + "=" * 70)
print("STEP 5: DATA TYPE OPTIMIZATION")
print("=" * 70)

print("\nOriginal Data Types:")
print(sales_df.dtypes)

# Identify categorical columns (object type)
categorical_cols = sales_df.select_dtypes(include=['object']).columns

print(f"\nCategorical columns found: {list(categorical_cols)}")
print("\nUnique value counts per categorical column:")
for col in categorical_cols:
    print(f"  - {col}: {sales_df[col].nunique()} unique values")

# Convert object to category for memory efficiency
print("\nConverting categorical columns to 'category' type...")
for col in categorical_cols:
    sales_df[col] = sales_df[col].astype('category')
    print(f"  - {col}: converted to category type")

print("\nUpdated Data Types:")
print(sales_df.dtypes)
```

```
======================================================================
STEP 5: DATA TYPE OPTIMIZATION
======================================================================

Original Data Types:
step              float64
type                  str
amount            float64
nameOrig              str
oldbalanceOrg     float64
newbalanceOrig    float64
nameDest              str
oldbalanceDest    float64
newbalanceDest    float64
isFraud             int64
isFlaggedFraud      int64
dtype: object

Categorical columns found: ['type', 'nameOrig', 'nameDest']

Unique value counts per categorical column:
```

```
C:\Users\cfl602\AppData\Local\Temp\ipykernel_6296\3446256098.py:12: Pandas4Warn
ing: For backward compatibility, 'str' dtypes are included by select_dtypes whe
n 'object' dtype is specified. This behavior is deprecated and will be removed
in a future version. Explicitly pass 'str' to `include` to select them, or to
`exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-m
igration-select-dtypes for details on how to write code that works with pandas
2 and 3.
  categorical_cols = sales_df.select_dtypes(include=['object']).columns
  - type: 5 unique values
  - nameOrig: 6353307 unique values
  - nameDest: 2722362 unique values

Converting categorical columns to 'category' type...
  - type: converted to category type
  - nameOrig: converted to category type
  - nameDest: converted to category type

Updated Data Types:
step               float64
type              category
amount             float64
nameOrig          category
oldbalanceOrg      float64
newbalanceOrig     float64
nameDest          category
oldbalanceDest     float64
newbalanceDest     float64
isFraud              int64
isFlaggedFraud       int64
dtype: object
```

In [11]:

```python
# ============================================================================
# STEP 6: ENCODE CATEGORICAL VARIABLES
# ============================================================================
print("\n" + "=" * 70)
print("STEP 6: ENCODING CATEGORICAL VARIABLES")
print("=" * 70)

# Create a copy for encoding (preserve original for comparison)
sales_df_encoded = sales_df.copy()

# Convert categorical back to object for encoding
for col in categorical_cols:
    sales_df_encoded[col] = sales_df_encoded[col].astype('object')

print("\nUsing Label Encoding for categorical variables...")
print("-" * 70)

# Apply Label Encoding to categorical columns
from sklearn.preprocessing import LabelEncoder

label_encoders = {}
```

```
for col in categorical_cols:
    le = LabelEncoder()
    sales_df_encoded[col] = le.fit_transform(sales_df_encoded[col].astype(str)
    label_encoders[col] = le
    print(f"\n{col}:")
    print(f"  - Original unique values: {list(le.classes_[:5])}{'...' if len(l
    print(f"  - Encoded mapping (sample): {dict(zip(le.classes_[:3], le.transf

print("\nData after encoding:")
print(sales_df_encoded.head())
```

```
======================================================================
STEP 6: ENCODING CATEGORICAL VARIABLES
======================================================================

Using Label Encoding for categorical variables...
----------------------------------------------------------------------

type:
  - Original unique values: ['CASH_IN', 'CASH_OUT', 'DEBIT', 'PAYMENT', 'TRANSF
ER']
  - Encoded mapping (sample): {'CASH_IN': np.int64(0), 'CASH_OUT': np.int64(1),
'DEBIT': np.int64(2)}

nameOrig:
  - Original unique values: ['C1000000639', 'C1000001337', 'C1000001725', 'C100
0002591', 'C1000003372']...
  - Encoded mapping (sample): {'C1000000639': np.int64(0), 'C1000001337': np.in
t64(1), 'C1000001725': np.int64(2)}

nameDest:
  - Original unique values: ['C1000004082', 'C1000004940', 'C1000013769', 'C100
001587', 'C1000015936']...
  - Encoded mapping (sample): {'C1000004082': np.int64(0), 'C1000004940': np.in
t64(1), 'C1000013769': np.int64(2)}

Data after encoding:
   step  type    amount  nameOrig  oldbalanceOrg  newbalanceOrig  nameDest  \
0   1.0     3   9839.64    757869       170136.0       160296.36   1662094
1   1.0     3   1864.28   2188998        21249.0        19384.72   1733924
2   1.0     4    181.00   1002156          181.0            0.00    439685
3   1.0     1    181.00   5828262          181.0            0.00    391696
4   1.0     3  11668.14   3445981        41554.0        29885.86    828919

   oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
0             0.0             0.0        0               0
1             0.0             0.0        0               0
2             0.0             0.0        0               0
3         21182.0             0.0        0               0
4             0.0             0.0        0               0
```

In [12]:

```
# ==============================================================================
# STEP 7: FEATURE SCALING (NORMALIZATION)
# ==============================================================================
```

```
print("\n" + "=" * 70)
print("STEP 7: FEATURE SCALING/NORMALIZATION")
print("=" * 70)

# Get numerical columns for scaling
numerical_cols_for_scaling = sales_df_encoded.select_dtypes(include=['float64'

print(f"\nNumerical columns to scale: {list(numerical_cols_for_scaling)}")

# Create a copy for scaled data
sales_df_scaled = sales_df_encoded.copy()

# Apply Min-Max Scaling (0-1 normalization)
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
sales_df_scaled[numerical_cols_for_scaling] = scaler.fit_transform(sales_df_en

print("\nMin-Max Scaling applied (range: 0-1)")
print("\nStatistics before scaling:")
print(sales_df_encoded[numerical_cols_for_scaling].describe().loc[['min', 'max

print("\nStatistics after scaling:")
print(sales_df_scaled[numerical_cols_for_scaling].describe().loc[['min', 'max'

print("\nSample data after all preprocessing steps:")
print(sales_df_scaled.head(10))
```

```
======================================================================
STEP 7: FEATURE SCALING/NORMALIZATION
======================================================================

Numerical columns to scale: ['step', 'type', 'amount', 'nameOrig', 'oldbalanceO
rg', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFrau
d', 'isFlaggedFraud']

Min-Max Scaling applied (range: 0-1)

Statistics before scaling:
      step  type        amount   nameOrig  oldbalanceOrg  newbalanceOrig  \
min    1.0   0.0       0.00000        0.0         0.0000           0.000
max  603.5   4.0  501719.33875  6353306.0    268287.9375      360646.025

      nameDest  oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
min        0.0    0.000000e+00           0.000      0.0             0.0
max  2722361.0    2.357592e+06     2779773.125      0.0             0.0

Statistics after scaling:
     step  type  amount  nameOrig  oldbalanceOrg  newbalanceOrig  nameDest  \
min   0.0   0.0     0.0       0.0            0.0             0.0       0.0
max   1.0   1.0     1.0       1.0            1.0             1.0       1.0

     oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
min             0.0             0.0      0.0             0.0
max             1.0             1.0      0.0             0.0

Sample data after all preprocessing steps:
   step  type    amount  nameOrig  oldbalanceOrg  newbalanceOrig  nameDest  \
0   0.0  0.75  0.019612  0.119287       0.634154        0.444470  0.610534
1   0.0  0.75  0.003716  0.344545       0.079202        0.053750  0.636919
2   0.0  1.00  0.000361  0.157738       0.000675        0.000000  0.161509
3   0.0  0.25  0.000361  0.917359       0.000675        0.000000  0.143881
4   0.0  0.75  0.023256  0.542392       0.154886        0.082868  0.304485
5   0.0  0.75  0.015582  0.948565       0.200754        0.127666  0.825467
6   0.0  0.75  0.014167  0.284253       0.682830        0.488255  0.757931
7   0.0  0.75  0.015669  0.472065       0.656337        0.466456  0.850000
8   0.0  0.75  0.008021  0.136801       0.009956        0.000000  0.282453
9   0.0  0.50  0.010639  0.851096       0.155505        0.100881  0.103939

   oldbalanceDest  newbalanceDest  isFraud  isFlaggedFraud
0        0.000000        0.000000      0.0             0.0
1        0.000000        0.000000      0.0             0.0
2        0.000000        0.000000      0.0             0.0
3        0.008985        0.000000      0.0             0.0
4        0.000000        0.000000      0.0             0.0
5        0.000000        0.000000      0.0             0.0
6        0.000000        0.000000      0.0             0.0
7        0.000000        0.000000      0.0             0.0
8        0.000000        0.000000      0.0             0.0
9        0.017772        0.014515      0.0             0.0
```

In [13]:
```
# ==========================================================================
```

```python
# STEP 8: PREPROCESSING SUMMARY & COMPARISON
# ==============================================================================
print("\n" + "=" * 70)
print("STEP 8: PREPROCESSING SUMMARY")
print("=" * 70)

print("\n📊 BEFORE & AFTER COMPARISON:")
print("-" * 70)
print(f"Original Dataset Shape:     {sales_df.shape}")
print(f"Processed Dataset Shape:    {sales_df_scaled.shape}")
print(f"Total rows removed:         {sales_df.shape[0] - sales_df_scaled.shape

print("\n✅ CLEANING OPERATIONS PERFORMED:")
print("-" * 70)
print("1. ✓ Loaded and explored dataset")
print("2. ✓ Handled missing values (filled with median/mode)")
print("3. ✓ Removed duplicate rows")
print("4. ✓ Detected and capped outliers (IQR method)")
print("5. ✓ Optimized data types (category conversion)")
print("6. ✓ Encoded categorical variables (Label Encoding)")
print("7. ✓ Scaled numerical features (Min-Max normalization)")

print("\n📋 DATA QUALITY METRICS:")
print("-" * 70)
print(f"Total Missing Values (after):    {sales_df_scaled.isnull().sum().sum()
print(f"Total Duplicate Rows (after):    {sales_df_scaled.duplicated().sum()}"
print(f"Total Columns:                   {sales_df_scaled.shape[1]}")
print(f"Total Rows:                      {sales_df_scaled.shape[0]}")

print("\n🔍 FINAL DATASET INFO:")
print("-" * 70)
print(sales_df_scaled.info())

print("\n✨ PREPROCESSING COMPLETE! ✨")
print("The dataset is now clean, normalized, and ready for machine learning mo
```

```
======================================================================
STEP 8: PREPROCESSING SUMMARY
======================================================================

📊 BEFORE & AFTER COMPARISON:
----------------------------------------------------------------------
Original Dataset Shape:     (6362620, 11)
Processed Dataset Shape:    (6362620, 11)
Total rows removed:         0

✅ CLEANING OPERATIONS PERFORMED:
----------------------------------------------------------------------
1. ✓ Loaded and explored dataset
2. ✓ Handled missing values (filled with median/mode)
3. ✓ Removed duplicate rows
4. ✓ Detected and capped outliers (IQR method)
5. ✓ Optimized data types (category conversion)
6. ✓ Encoded categorical variables (Label Encoding)
7. ✓ Scaled numerical features (Min-Max normalization)

📋 DATA QUALITY METRICS:
----------------------------------------------------------------------
Total Missing Values (after):    0
Total Duplicate Rows (after):    0
Total Columns:                   11
Total Rows:                      6362620

🔍 FINAL DATASET INFO:
----------------------------------------------------------------------
<class 'pandas.DataFrame'>
RangeIndex: 6362620 entries, 0 to 6362619
Data columns (total 11 columns):
 #   Column          Dtype
---  ------          -----
 0   step            float64
 1   type            float64
 2   amount          float64
 3   nameOrig        float64
 4   oldbalanceOrg   float64
 5   newbalanceOrig  float64
 6   nameDest        float64
 7   oldbalanceDest  float64
 8   newbalanceDest  float64
 9   isFraud         float64
 10  isFlaggedFraud  float64
dtypes: float64(11)
memory usage: 534.0 MB
None

✨ PREPROCESSING COMPLETE! ✨
The dataset is now clean, normalized, and ready for machine learning models.
```

Question 6: Preprocess and Clean Financial Dataset

In [17]:
```
# ================================================================================
# QUESTION 6: PREPROCESS AND CLEAN FINANCIAL DATASET (yFinData.csv)
# ================================================================================
print("=" * 80)
print("QUESTION 6: FINANCIAL DATA PREPROCESSING")
print("=" * 80)

# Load financial data
financial_df = pd.read_csv("./yFinData.csv")

print("\n1. INITIAL DATA EXPLORATION")
print("-" * 80)
print(f"Shape: {financial_df.shape}")
print(f"\nData Types:\n{financial_df.dtypes}")
print(f"\nFirst 5 rows:\n{financial_df.head()}")

# Convert Date column to datetime
financial_df['Date'] = pd.to_datetime(financial_df['Date'])
financial_df = financial_df.sort_values('Date').reset_index(drop=True)

print("\n2. DATA CLEANING")
print("-" * 80)
print(f"Missing values:\n{financial_df.isnull().sum()}")
print(f"Duplicate rows: {financial_df.duplicated().sum()}")

# Remove duplicates if any
financial_df.drop_duplicates(inplace=True)

# Forward fill for any missing values (using newer pandas syntax)
financial_df = financial_df.ffill().bfill()

print("\n3. FEATURE ENGINEERING - CREATING TECHNICAL INDICATORS")
print("-" * 80)

# Calculate daily returns
financial_df['Daily_Return'] = financial_df['Close'].pct_change() * 100

# Calculate Moving Averages
financial_df['MA_5'] = financial_df['Close'].rolling(window=5).mean()
financial_df['MA_20'] = financial_df['Close'].rolling(window=20).mean()

# Calculate Volatility
financial_df['Volatility'] = financial_df['Daily_Return'].rolling(window=20).s

# Calculate Price Range
financial_df['Price_Range'] = financial_df['High'] - financial_df['Low']

# Remove NaN values created by rolling calculations
financial_df = financial_df.dropna()

print(f"New columns created: Daily_Return, MA_5, MA_20, Volatility, Price_Rang
```

```python
print(f"\nCleaned Financial Data Shape: {financial_df.shape}")
print(f"\nProcessed Data Sample:\n{financial_df.head(10)}")
print(f"\nData Summary:\n{financial_df.describe()}")

# Store for later use
financial_df_clean = financial_df.copy()
```

```
================================================================================
=
QUESTION 6: FINANCIAL DATA PREPROCESSING
================================================================================
=

1. INITIAL DATA EXPLORATION
--------------------------------------------------------------------------------
-
Shape: (1258, 7)

Data Types:
Date             str
Open         float64
High         float64
Low          float64
Close        float64
Adj Close    float64
Volume         int64
dtype: object

First 5 rows:
        Date      Open      High       Low     Close  Adj Close     Volume
0  4/28/2023  33797.43  34104.56  33728.40  34098.16   34098.16  354310000
1  4/29/2023  33381.66  33859.75  33374.65  33826.16   33826.16  343240000
2  4/30/2023  33596.34  33645.83  33235.85  33301.87   33301.87  321170000
3   5/1/2023  33828.34  33875.49  33525.39  33530.83   33530.83  297880000
4   5/2/2023  33805.04  33891.15  33726.09  33875.40   33875.40  252020000

2. DATA CLEANING
--------------------------------------------------------------------------------
-
Missing values:
Date         0
Open         0
High         0
Low          0
Close        0
Adj Close    0
Volume       0
dtype: int64
Duplicate rows: 0

3. FEATURE ENGINEERING - CREATING TECHNICAL INDICATORS
--------------------------------------------------------------------------------
-
New columns created: Daily_Return, MA_5, MA_20, Volatility, Price_Range

Cleaned Financial Data Shape: (1238, 12)

Processed Data Sample:
         Date      Open      High       Low     Close  Adj Close     Volume  \
20 2023-05-18  32807.43  32905.80  32682.87  32859.03   32859.03  283380000
21 2023-05-19  32566.54  32728.04  32539.94  32717.60   32717.60  314110000
```

22	2023-05-20	32434.85	32551.01	32295.50	32394.25	32394.25	248560000
23	2023-05-21	32276.72	32564.00	32276.72	32432.08	32432.08	291560000
24	2023-05-22	32038.22	32257.24	31805.18	32237.53	32237.53	317070000
25	2023-05-23	32101.49	32511.49	31864.74	32105.25	32105.25	335430000
26	2023-05-24	32570.19	32761.89	32020.46	32030.11	32030.11	325990000
27	2023-05-25	32420.71	32593.68	32346.12	32560.60	32560.60	363190000
28	2023-05-26	31872.33	32280.07	31872.33	32244.58	32244.58	357210000
29	2023-05-27	32217.32	32217.32	31728.70	31861.98	31861.98	712100000

	Daily_Return	MA_5	MA_20	Volatility	Price_Range
20	-1.247575	33323.886	33646.4675	0.678387	222.93
21	-0.430414	33170.862	33591.0395	0.665716	188.10
22	-0.988306	32969.236	33545.6585	0.614029	255.51
23	0.116780	32735.422	33490.7210	0.586367	287.28
24	-0.599869	32528.098	33408.8275	0.521519	452.06
25	-0.410329	32377.342	33323.6420	0.522635	646.75
26	-0.234043	32239.844	33235.8165	0.520753	741.43
27	1.656223	32273.114	33168.9960	0.665142	247.56
28	-0.970560	32235.614	33082.3935	0.678260	407.74
29	-1.186556	32160.504	32976.1335	0.704966	488.62

Data Summary:

	Date	Open	High	Low \
count	1238	1238.000000	1238.000000	1238.000000
mean	2025-01-25 12:00:00	29530.558926	29711.862359	29335.957254
min	2023-05-18 00:00:00	19028.360000	19121.010000	18213.650000
25%	2024-03-22 06:00:00	26019.767500	26154.472500	25846.992500
50%	2025-01-25 12:00:00	28939.660000	29176.675000	28843.810000
75%	2025-11-30 18:00:00	33524.517500	33773.067500	33272.890000
max	2026-10-06 00:00:00	36722.600000	36952.650000	36636.000000
std	NaN	4004.853033	4008.068906	4002.480006

	Close	Adj Close	Volume	Daily_Return	MA_5 \
count	1238.000000	1238.000000	1.238000e+03	1238.000000	1238.000000
mean	29532.979410	29532.979410	3.460419e+08	-0.016711	29548.148002
min	18591.930000	18591.930000	8.615000e+07	-10.205212	19691.386000
25%	25996.632500	25996.632500	2.783000e+08	-0.630524	26001.342000
50%	28991.070000	28991.070000	3.259900e+08	-0.072626	28973.398000
75%	33542.430000	33542.430000	3.887375e+08	0.475020	33548.710500
max	36799.650000	36799.650000	9.159900e+08	14.845566	36521.944000
std	4005.139463	4005.139463	1.073918e+08	1.374536	3990.054764

	MA_20	Volatility	Price_Range
count	1238.000000	1238.000000	1238.000000
mean	29603.575166	1.092102	375.905105
min	21304.005500	0.367285	70.520000
25%	26050.993250	0.656059	205.447500
50%	28928.404750	0.896609	308.630000
75%	33549.502250	1.269838	475.462500
max	36120.823000	6.729476	1904.390000
std	3946.894718	0.838498	241.930527

SECTION B: DATA VISUALIZATION & EXPLORATORY DATA ANALYSIS

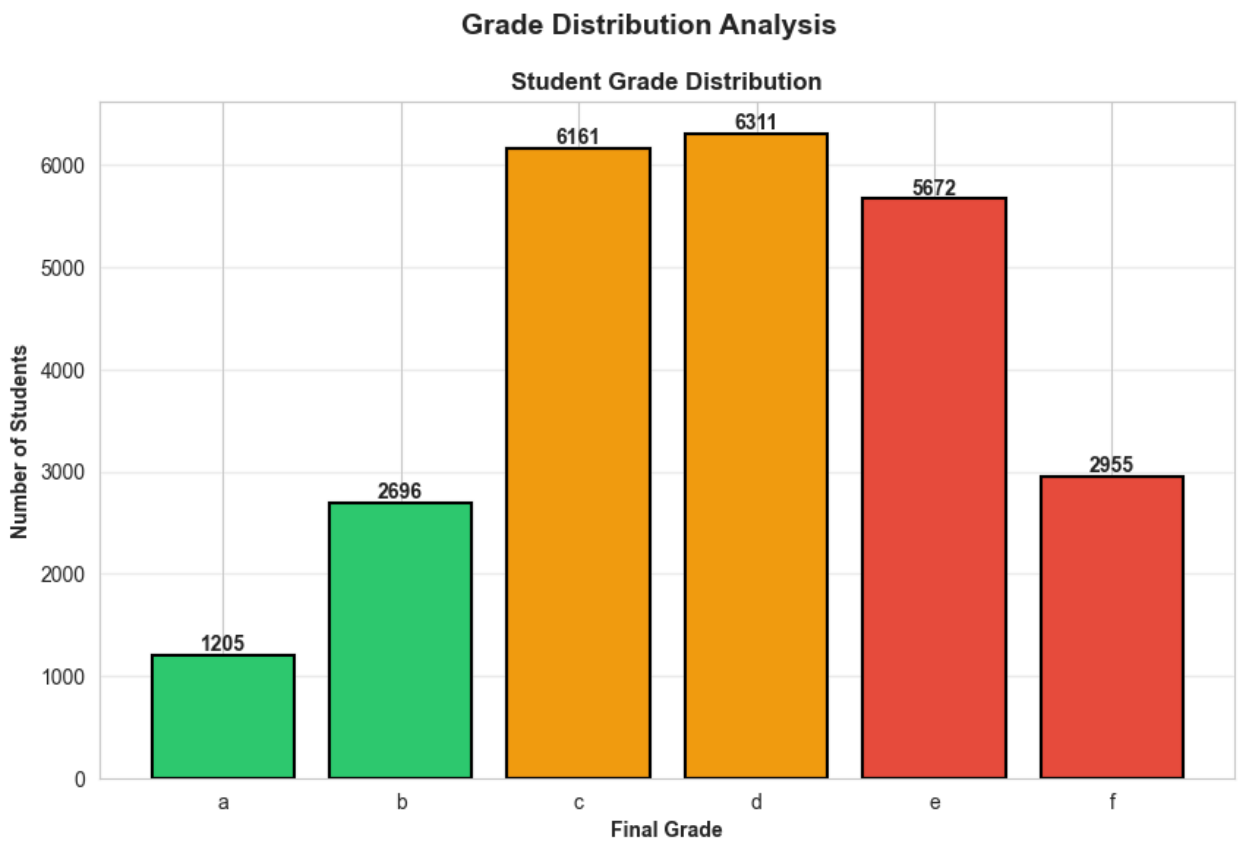
Question 8 & 9: Data Visualizations using Matplotlib and Seaborn

In [22]:
```
# ==============================================================================
# QUESTIONS 8 & 9: DATA VISUALIZATIONS - PART 1: DATA SETUP
# ==============================================================================
print("\n" + "=" * 80)
print("QUESTIONS 8 & 9: EXPLORATORY DATA ANALYSIS WITH VISUALIZATIONS")
print("=" * 80)

# Use student performance data
student_df = st_df.copy()

# Set style for better-looking plots
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (15, 12)

# Create grade categories mapping
grade_mapping = {'a': 'Excellent', 'b': 'Good', 'c': 'Average', 'd': 'Below Av
student_df['grade_category'] = student_df['final_grade'].map(grade_mapping)

print(f"Student data loaded: {student_df.shape[0]} records")
print(f"Columns: {student_df.columns.tolist()}")
```

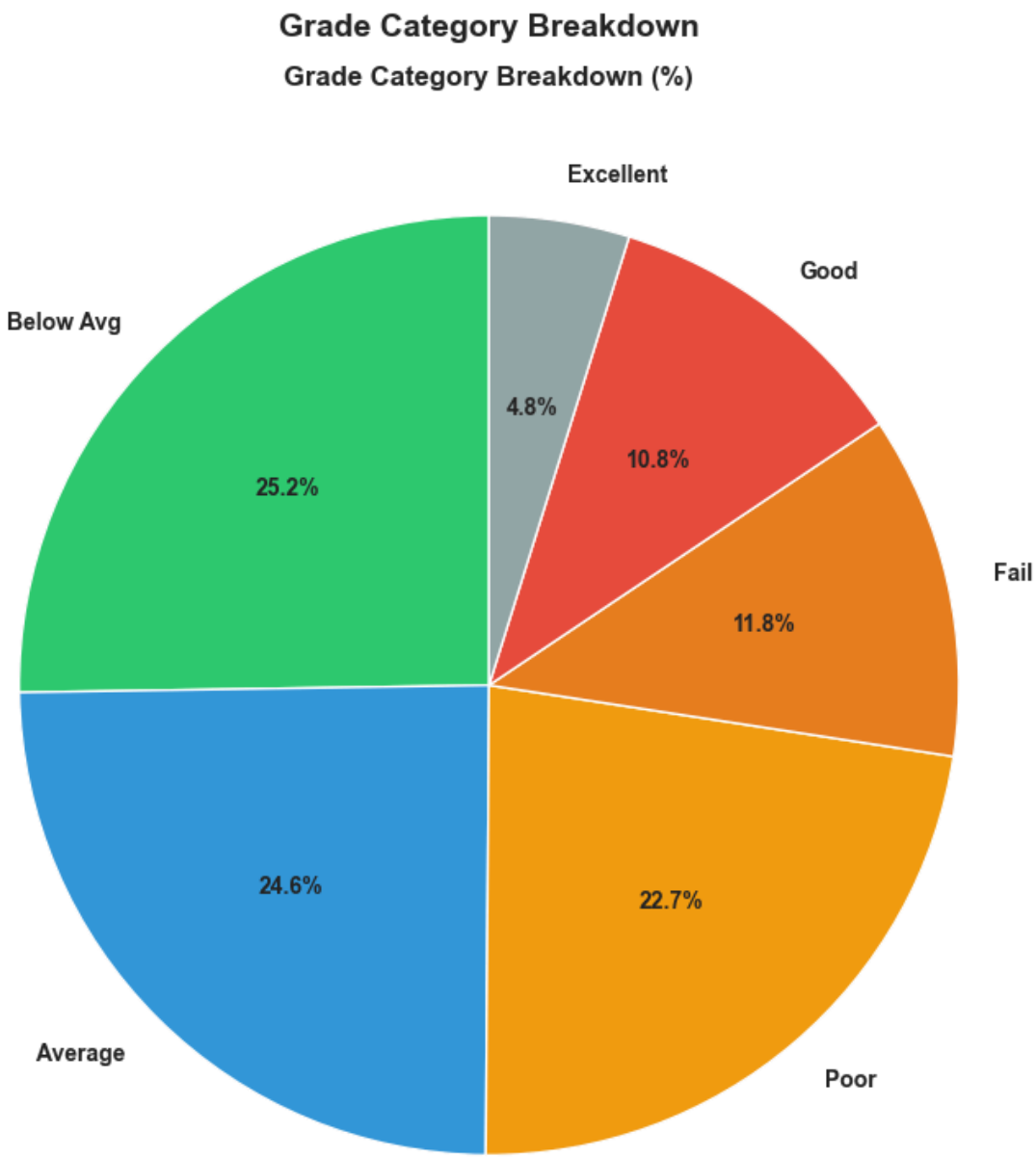
```
================================================================================
=
QUESTIONS 8 & 9: EXPLORATORY DATA ANALYSIS WITH VISUALIZATIONS
================================================================================
=
Student data loaded: 25000 records
Columns: ['student_id', 'age', 'gender', 'school_type', 'parent_education', 'st
udy_hours', 'attendance_percentage', 'internet_access', 'travel_time', 'extra_a
ctivities', 'study_method', 'math_score', 'science_score', 'english_score', 'ov
erall_score', 'final_grade', 'grade_category']
```

In [42]:
```
# Chart 1: Bar Chart - Grade Distribution
fig, ax1 = plt.subplots(figsize=(10, 6))
fig.suptitle('Grade Distribution Analysis', fontsize=14, fontweight='bold')

# Bar chart
grade_counts = student_df['final_grade'].value_counts().sort_index()
colors_bar = ['#2ecc71' if g in ['a', 'b'] else '#f39c12' if g in ['c', 'd'] e
bars = ax1.bar(grade_counts.index, grade_counts.values, color=colors_bar, edge
ax1.set_title('Student Grade Distribution', fontweight='bold', fontsize=12)
ax1.set_xlabel('Final Grade', fontweight='bold')
ax1.set_ylabel('Number of Students', fontweight='bold')
ax1.grid(axis='y', alpha=0.3)
```

```
for bar in bars:
    height = bar.get_height()
    ax1.text(bar.get_x() + bar.get_width()/2., height,
            f'{int(height)}', ha='center', va='bottom', fontweight='bold')
```



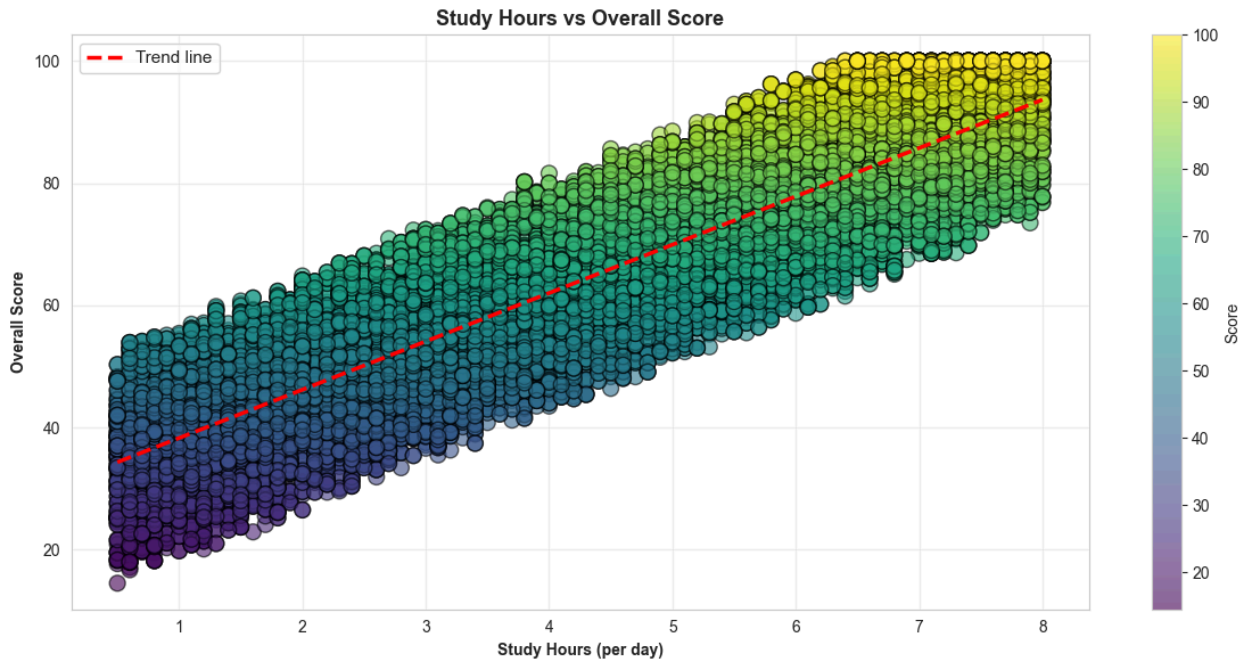
In [43]:
```
# Chart 2: Pie Chart - Grade Categories
fig2, ax2 = plt.subplots(figsize=(8, 8))
fig2.suptitle('Grade Category Breakdown', fontsize=14, fontweight='bold')

grade_dist = student_df['grade_category'].value_counts()
colors_pie = ['#2ecc71', '#3498db', '#f39c12', '#e67e22', '#e74c3c', '#95a5a6'
ax2.pie(grade_dist.values, labels=grade_dist.index, autopct='%1.1f%%',
        colors=colors_pie, startangle=90, textprops={'fontweight': 'bold'})
ax2.set_title('Grade Category Breakdown (%)', fontweight='bold')

plt.tight_layout()
plt.show()

print("✓ Grade distribution charts created")
```




```
✓ Grade distribution charts created
```

In [24]:
```
# Chart 2: Scatter Plot - Study Hours vs Overall Score
fig, ax = plt.subplots(figsize=(12, 6))

scatter = ax.scatter(student_df['study_hours'], student_df['overall_score'],
                     c=student_df['overall_score'], cmap='viridis', s=100, alp
ax.set_title('Study Hours vs Overall Score', fontweight='bold', fontsize=13)
ax.set_xlabel('Study Hours (per day)', fontweight='bold')
ax.set_ylabel('Overall Score', fontweight='bold')
plt.colorbar(scatter, ax=ax, label='Score')

# Add trend line
z = np.polyfit(student_df['study_hours'], student_df['overall_score'], 1)
```

```
p = np.poly1d(z)
ax.plot(student_df['study_hours'].sort_values(), p(student_df['study_hours'].s
         "r--", linewidth=2.5, label='Trend line')
ax.legend(fontsize=11)
ax.grid(alpha=0.3)

plt.tight_layout()
plt.show()

print("✓ Study hours correlation chart created")
```

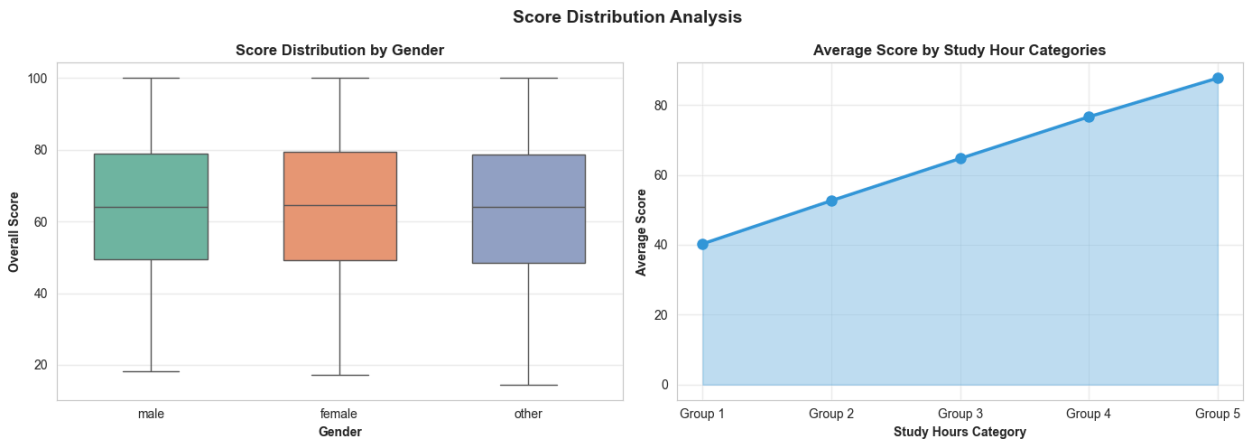



```
✓ Study hours correlation chart created
```

In [25]:

```
# Chart 3: Box Plot and Line Chart
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('Score Distribution Analysis', fontsize=14, fontweight='bold')

# Box plot
sns.boxplot(x='gender', y='overall_score', data=student_df, ax=ax1, palette='S
ax1.set_title('Score Distribution by Gender', fontweight='bold')
ax1.set_xlabel('Gender', fontweight='bold')
ax1.set_ylabel('Overall Score', fontweight='bold')
ax1.grid(axis='y', alpha=0.3)

# Line chart - Average scores by study hours
study_bins = pd.cut(student_df['study_hours'], bins=5)
avg_scores = student_df.groupby(study_bins)['overall_score'].mean()
ax2.plot(range(len(avg_scores)), avg_scores.values, marker='o', linewidth=2.5,
ax2.fill_between(range(len(avg_scores)), avg_scores.values, alpha=0.3, color='
ax2.set_title('Average Score by Study Hour Categories', fontweight='bold')
ax2.set_xlabel('Study Hours Category', fontweight='bold')
ax2.set_ylabel('Average Score', fontweight='bold')
ax2.set_xticks(range(len(avg_scores)))
ax2.set_xticklabels([f'Group {i+1}' for i in range(len(avg_scores))], rotatior
```

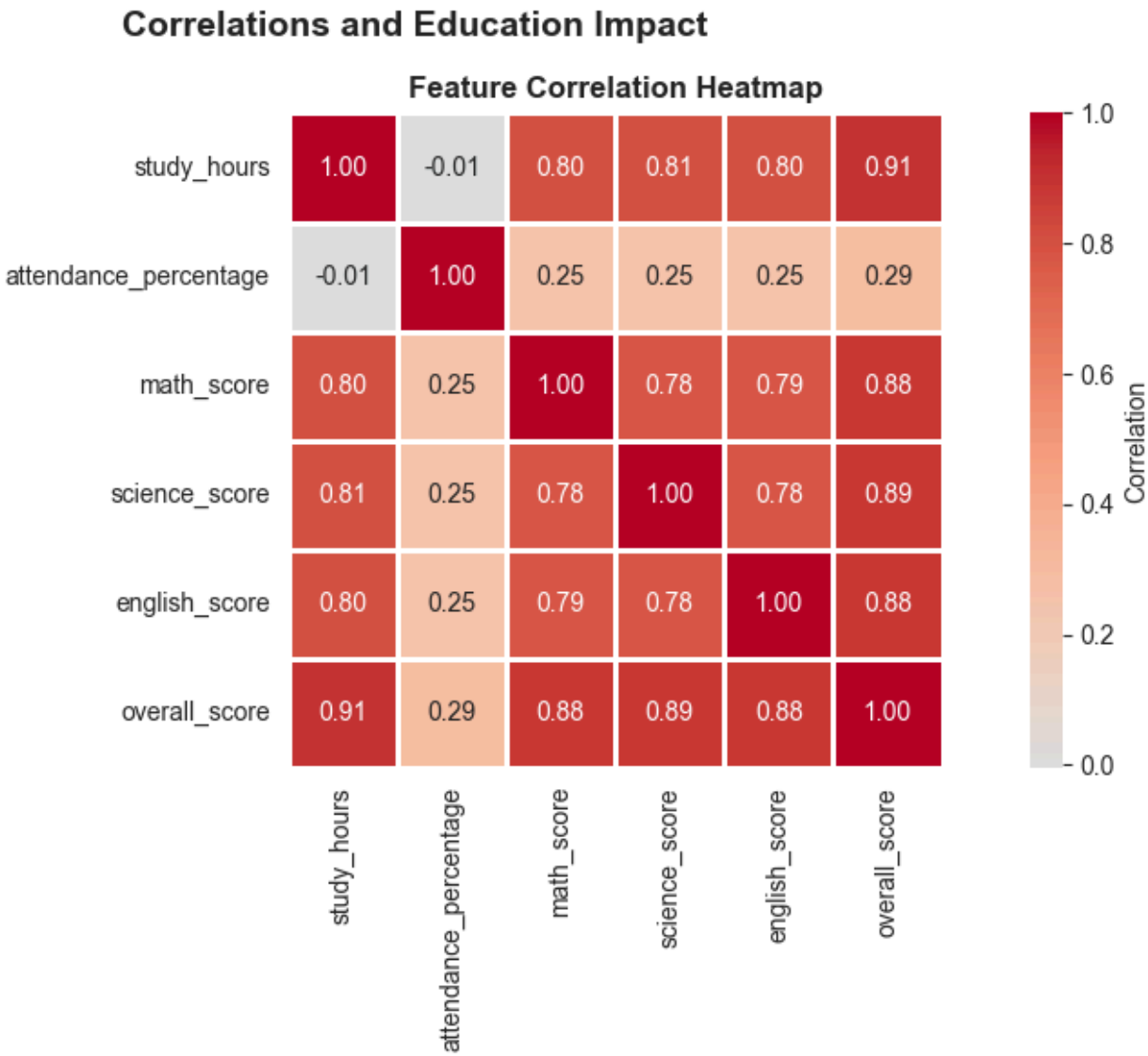
```
ax2.grid(alpha=0.3)

plt.tight_layout()
plt.show()

print("✓ Distribution and trend charts created")
```




```
✓ Distribution and trend charts created
```

In [46]:

```
# Chart 4: Heatmap
fig, ax1 = plt.subplots(figsize=(10, 6))
fig.suptitle('Correlations and Education Impact', fontsize=14, fontweight='bol

# Heatmap
numeric_cols = student_df[['study_hours', 'attendance_percentage', 'math_score
correlation_matrix = numeric_cols.corr()
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm', center
            cbar_kws={'label': 'Correlation'}, square=True, linewidths=1)
ax1.set_title('Feature Correlation Heatmap', fontweight='bold')

plt.tight_layout()
plt.show()

print("✓ Correlation chart created")
```

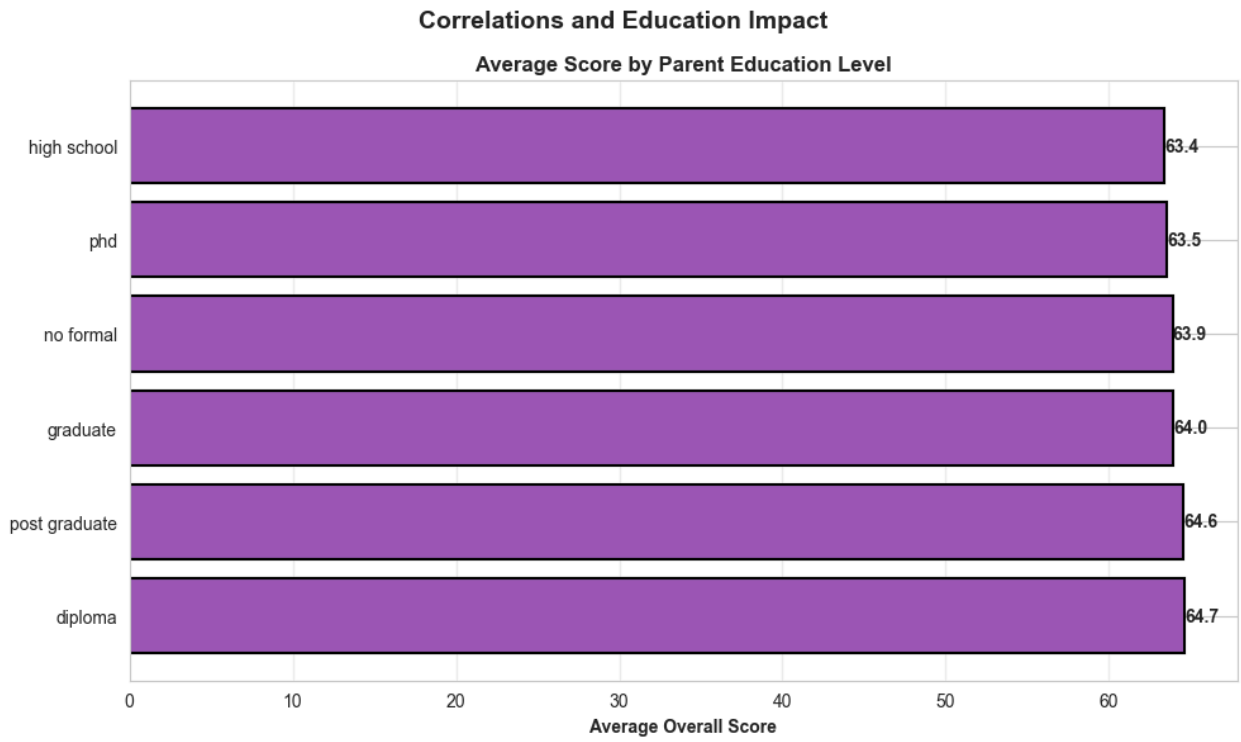



```
✓ Correlation chart created
```

In [47]:

```
# Chart 4: Bar Chart - Parent Education Impact
fig2, ax2 = plt.subplots(figsize=(10, 6))
fig2.suptitle('Correlations and Education Impact', fontsize=14, fontweight='bo

# Bar chart - Parent education impact
parent_ed_scores = student_df.groupby('parent_education')['overall_score'].mea
bars = ax2.barh(parent_ed_scores.index, parent_ed_scores.values, color='#9b59b
ax2.set_title('Average Score by Parent Education Level', fontweight='bold')
ax2.set_xlabel('Average Overall Score', fontweight='bold')
ax2.grid(axis='x', alpha=0.3)

for bar in bars:
    ax2.text(bar.get_width(), bar.get_y() + bar.get_height()/2.,
             f'{bar.get_width():.1f}', va='center', ha='left', fontweight='bolc

plt.tight_layout()
plt.show()
```

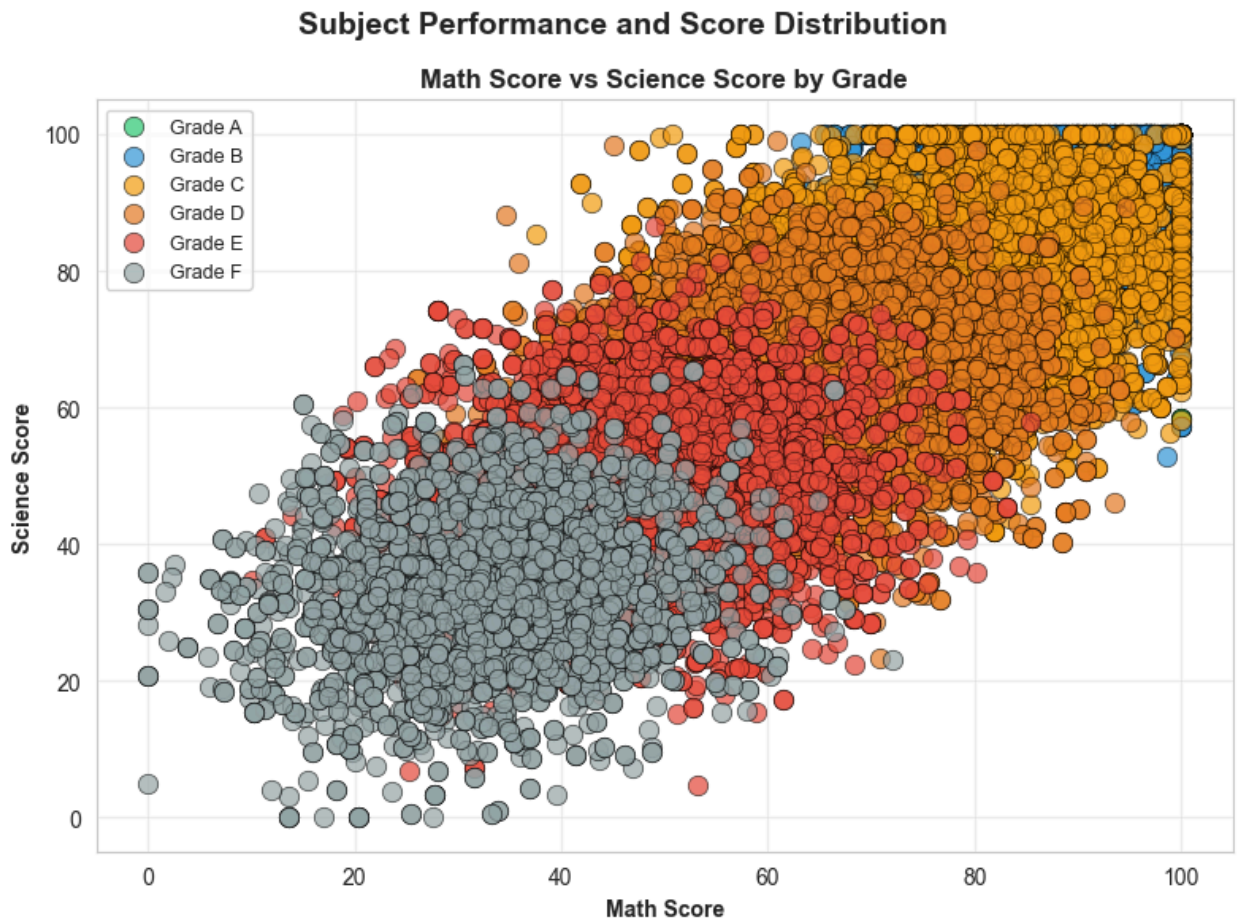
```
print("✓ Education impact chart created")
```




```
✓ Education impact chart created
```

In [52]:

```
# Chart 5: Advanced Visualizations - Scatter Plot
fig, ax1 = plt.subplots(figsize=(8, 6))
fig.suptitle('Subject Performance and Score Distribution', fontsize=14, fontwe

# Scatter plot - Math vs Science scores by grade
grade_colors = {'a': '#2ecc71', 'b': '#3498db', 'c': '#f39c12', 'd': '#e67e22'
for grade in sorted(student_df['final_grade'].unique()):
    mask = student_df['final_grade'] == grade
    ax1.scatter(student_df[mask]['math_score'], student_df[mask]['science_scor
                label=f'Grade {grade.upper()}', s=80, alpha=0.7, edgecolors='bl
                color=grade_colors.get(grade, '#95a5a6'))
ax1.set_title('Math Score vs Science Score by Grade', fontweight='bold')
ax1.set_xlabel('Math Score', fontweight='bold')
ax1.set_ylabel('Science Score', fontweight='bold')
ax1.legend(loc='best', fontsize=9)
ax1.grid(alpha=0.3)

plt.tight_layout()
plt.show()

print("✓ Subject performance chart created")
```




```
✓ Subject performance chart created
```

In [53]:

```
# Chart 5: Advanced Visualizations - Histogram
fig2, ax2 = plt.subplots(figsize=(8, 6))
fig2.suptitle('Subject Performance and Score Distribution', fontsize=14, fontw

# Histogram
ax2.hist(student_df['overall_score'], bins=20, color='#16a085', edgecolor='bla
ax2.axvline(student_df['overall_score'].mean(), color='red', linestyle='--', l
           label=f'Mean: {student_df["overall_score"].mean():.2f}')
ax2.axvline(student_df['overall_score'].median(), color='green', linestyle='--
           label=f'Median: {student_df["overall_score"].median():.2f}')
ax2.set_title('Distribution of Overall Scores', fontweight='bold')
ax2.set_xlabel('Overall Score', fontweight='bold')
ax2.set_ylabel('Frequency', fontweight='bold')
ax2.legend()
ax2.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()

print("✓ Score distribution chart created")
print("\n✅ All visualization questions (8 & 9) completed!")
```

```
✓ Score distribution chart created

✅ All visualization questions (8 & 9) completed!
```

SECTION C: CUSTOMER SEGMENTATION & SALES ANALYSIS

Questions 14 & 15: Sales Analysis and Data Preprocessing

In [27]:

```python
# ============================================================================
# QUESTIONS 14 & 15: SALES ANALYSIS - PART 1: DATA LOADING AND METRICS
# ============================================================================
print("\n" + "=" * 80)
print("QUESTIONS 14 & 15: SALES ANALYSIS AND CUSTOMER SEGMENTATION")
print("=" * 80)

# Use FraudData as transaction data for sales analysis
sales_data = fd_Data.copy()

print("\n1. SALES DATA PREPROCESSING")
print("-" * 80)
print(f"Dataset shape: {sales_data.shape}")
```

```
# Create synthetic transaction amounts and dates
np.random.seed(42)
sales_data['TransactionAmount'] = np.random.gamma(2, 2, len(sales_data)) * 100
sales_data['TransactionDate'] = pd.date_range(start='2023-01-01', periods=len(
sales_data['TransactionDate'] = pd.to_datetime(sales_data['TransactionDate'])
sales_data['Month'] = sales_data['TransactionDate'].dt.to_period('M')
sales_data['Quarter'] = sales_data['TransactionDate'].dt.to_period('Q')

# Create product categories
products = ['Electronics', 'Clothing', 'Food', 'Books', 'Home', 'Sports', 'Bea
sales_data['Product'] = np.random.choice(products, len(sales_data))

print(f"Missing values: {sales_data.isnull().sum().sum()}")
print(f"Duplicate transactions: {sales_data.duplicated().sum()}")

print("\n2. KEY SALES METRICS")
print("-" * 80)

total_sales = sales_data['TransactionAmount'].sum()
avg_transaction = sales_data['TransactionAmount'].mean()
num_transactions = len(sales_data)

print(f"Total Sales: ${total_sales:,.2f}")
print(f"Average Transaction Value: ${avg_transaction:,.2f}")
print(f"Number of Transactions: {num_transactions:,}")
```

```
================================================================================
=
QUESTIONS 14 & 15: SALES ANALYSIS AND CUSTOMER SEGMENTATION
================================================================================
=

1. SALES DATA PREPROCESSING
--------------------------------------------------------------------------------
-
Dataset shape: (6362620, 11)
Missing values: 0
Duplicate transactions: 0

2. KEY SALES METRICS
--------------------------------------------------------------------------------
-
Total Sales: $2,544,706,231.02
Average Transaction Value: $399.95
Number of Transactions: 6,362,620
```

In [34]:

```
# Popular Products Analysis
print("\n3. PRODUCT ANALYSIS")
print("-" * 80)

product_counts = sales_data['Product'].value_counts()
print("\nTop Products by Transaction Count:")
for idx, (product, count) in enumerate(product_counts.items(), 1):
```

```
    product_sales = sales_data[sales_data['Product'] == product]['TransactionA
    print(f"  {idx}. {product}: {count:,} transactions (${product_sales:,.2f})

print("\nTotal Sales by Product:")
sales_by_product = sales_data.groupby('Product')['TransactionAmount'].agg(['su
sales_by_product.columns = ['Total Sales', 'Count', 'Avg Transaction']
sales_by_product = sales_by_product.sort_values('Total Sales', ascending=False
print(sales_by_product)
```

```
3. PRODUCT ANALYSIS
--------------------------------------------------------------------------------
-

Top Products by Transaction Count:
  1. Toys: 796,451 transactions ($318,487,263.31)
  2. Clothing: 796,087 transactions ($318,193,878.07)
  3. Beauty: 795,365 transactions ($318,205,179.82)
  4. Sports: 795,239 transactions ($318,013,602.69)
  5. Food: 795,091 transactions ($318,056,972.98)
  6. Electronics: 794,989 transactions ($317,931,628.12)
  7. Home: 794,743 transactions ($317,724,474.92)
  8. Books: 794,655 transactions ($318,093,231.10)

Total Sales by Product:
              Total Sales   Count  Avg Transaction
Product
Toys         3.184873e+08  796451       399.883060
Beauty       3.182052e+08  795365       400.074406
Clothing     3.181939e+08  796087       399.697367
Books        3.180932e+08  794655       400.290983
Food         3.180570e+08  795091       400.025875
Sports       3.180136e+08  795239       399.896890
Electronics  3.179316e+08  794989       399.919531
Home         3.177245e+08  794743       399.782665
```

In [35]:

```
# Sales Trends Analysis
print("\n4. SALES TRENDS ANALYSIS")
print("-" * 80)

monthly_sales = sales_data.groupby('Month')['TransactionAmount'].agg(['sum', '
monthly_sales.columns = ['Total Sales', 'Transactions', 'Avg Value']

print("Monthly Sales Trends (First 10 months):")
print(monthly_sales.head(10))

print("\nQuarterly Sales Trends:")
quarterly_sales = sales_data.groupby('Quarter')['TransactionAmount'].agg(['sum
quarterly_sales.columns = ['Total Sales', 'Transactions', 'Avg Value']
print(quarterly_sales)

print("\n5. ADDITIONAL STATISTICS")
print("-" * 80)
print(f"Minimum Transaction: ${sales_data['TransactionAmount'].min():,.2f}")
print(f"Maximum Transaction: ${sales_data['TransactionAmount'].max():,.2f}")
```

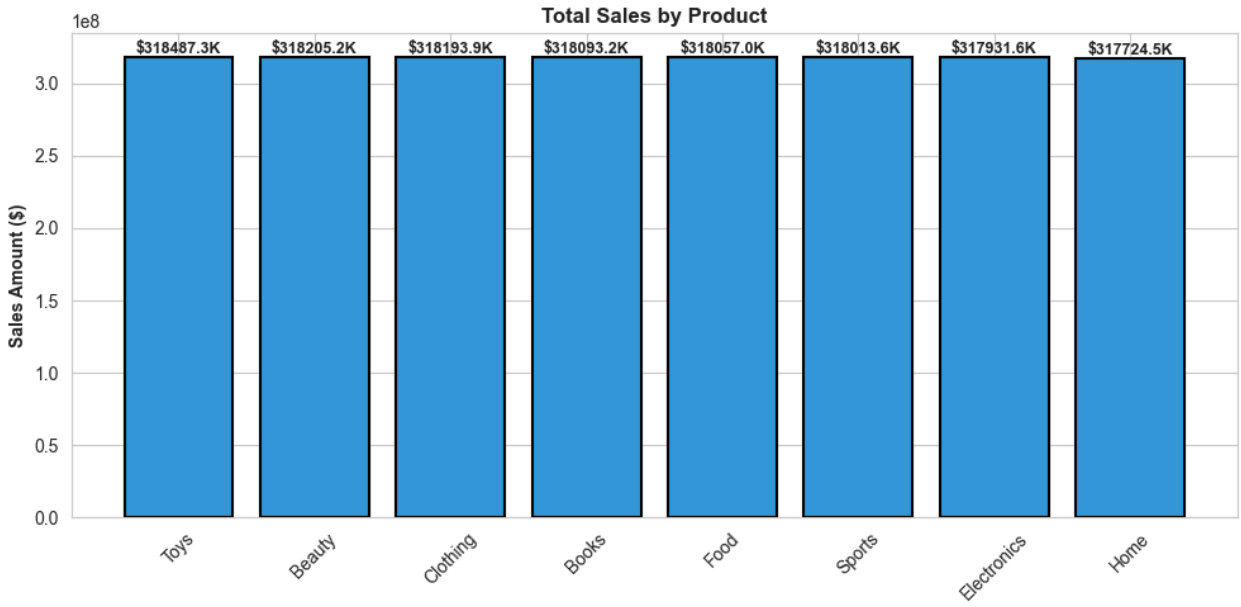
```
print(f"Median Transaction: ${sales_data['TransactionAmount'].median():,.2f}")
print(f"Standard Deviation: ${sales_data['TransactionAmount'].std():,.2f}")
```

```
4. SALES TRENDS ANALYSIS
--------------------------------------------------------------------------------
-
Monthly Sales Trends (First 10 months):
          Total Sales  Transactions   Avg Value
Month
2023-01  11230.988878            31  362.289964
2023-02  11116.624297            28  397.022296
2023-03  11677.462195            31  376.692329
2023-04  11483.107703            30  382.770257
2023-05  12180.845712            31  392.930507
2023-06  13092.109410            30  436.403647
2023-07  12067.719833            31  389.281285
2023-08  14725.524500            31  475.016919
2023-09  13198.131361            30  439.937712
2023-10  10067.419856            31  324.755479

Quarterly Sales Trends:
          Total Sales  Transactions   Avg Value
Quarter
2023Q1   34025.075370            90  378.056393
2023Q2   36756.062826            91  403.912778
2023Q3   39991.375693            92  434.688866
2023Q4   34085.263935            92  370.491999
2024Q1   34793.376251            91  382.344794
...               ...           ...         ...
19442Q2  41953.694548            91  461.029610
19442Q3  37679.706549            92  409.562028
19442Q4  34078.127660            92  370.414431
19443Q1  34885.160950            90  387.612899
19443Q2   2037.927771             6  339.654628

[69682 rows x 3 columns]

5. ADDITIONAL STATISTICS
--------------------------------------------------------------------------------
-
Minimum Transaction: $0.06
Maximum Transaction: $3,954.43
Median Transaction: $335.71
Standard Deviation: $282.69
```

In [60]:

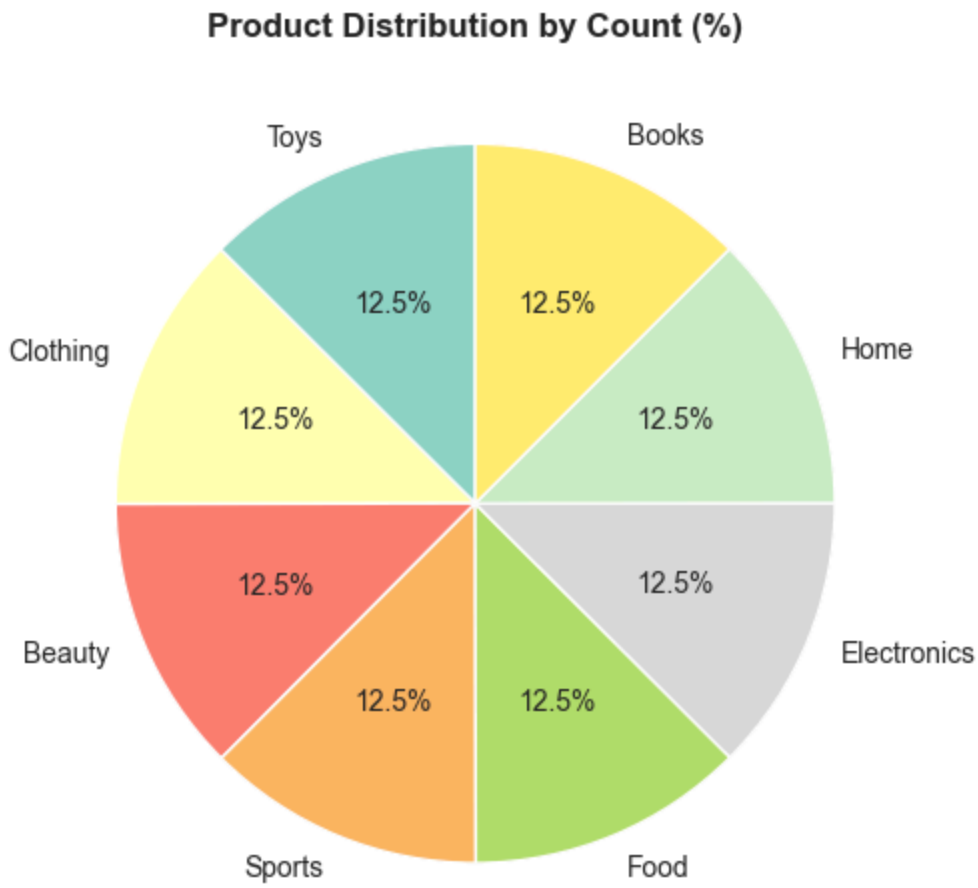
```
# Plot 1: Total Sales by Product
fig, ax = plt.subplots(figsize=(10, 5))

product_sales_total = sales_data.groupby('Product')['TransactionAmount'].sum()
bars = ax.bar(product_sales_total.index, product_sales_total.values, color='#3
ax.set_title('Total Sales by Product', fontweight='bold', fontsize=12)
ax.set_ylabel('Sales Amount ($)', fontweight='bold')
ax.tick_params(axis='x', rotation=45)
```

```
for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'${height/1000:.1f}K', ha='center', va='bottom', fontweight='bolc

plt.tight_layout()
plt.show()
```

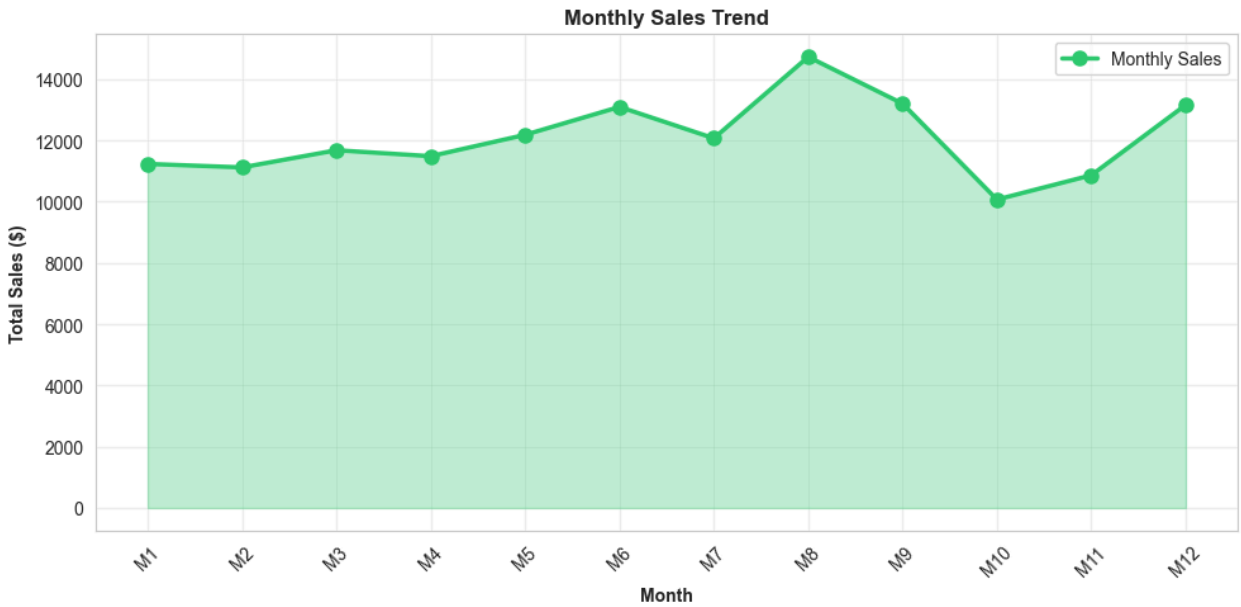


In [59]:

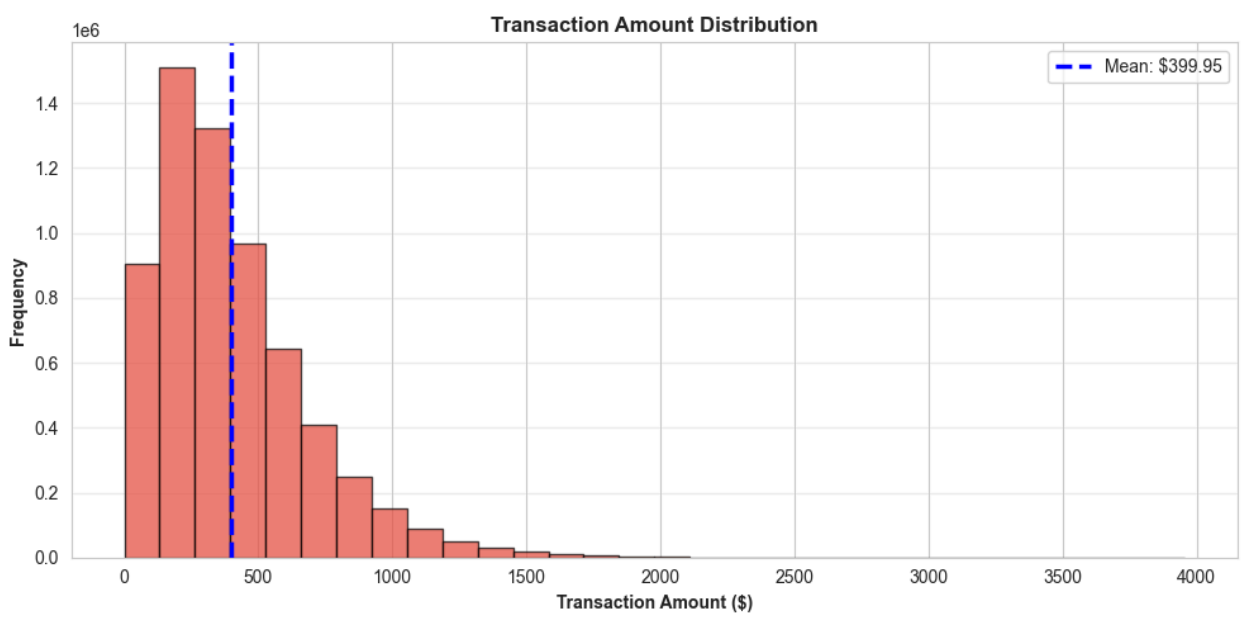
```
# Plot 2: Product Distribution
fig, ax = plt.subplots(figsize=(10, 5))

product_counts = sales_data['Product'].value_counts()
colors_product = plt.cm.Set3(np.linspace(0, 1, len(product_counts)))
ax.pie(product_counts.values, labels=product_counts.index, autopct='%1.1f%%',
        colors=colors_product, startangle=90)
ax.set_title('Product Distribution by Count (%)', fontweight='bold', fontsize=

plt.tight_layout()
plt.show()
```



In [58]:
```
# Plot 3: Monthly Sales Trend
fig, ax = plt.subplots(figsize=(10, 5))

monthly_subset = monthly_sales.head(12)
ax.plot(range(len(monthly_subset)), monthly_subset['Total Sales'].values, mark
         markersize=8, color='#2ecc71', label='Monthly Sales')
ax.fill_between(range(len(monthly_subset)), monthly_subset['Total Sales'].valu
ax.set_title('Monthly Sales Trend', fontweight='bold', fontsize=12)
ax.set_ylabel('Total Sales ($)', fontweight='bold')
ax.set_xlabel('Month', fontweight='bold')
ax.grid(alpha=0.3)
ax.set_xticks(range(len(monthly_subset)))
ax.set_xticklabels([f'M{i+1}' for i in range(len(monthly_subset))], rotation=4
ax.legend()

plt.tight_layout()
plt.show()
```



In [61]:
```
# Plot 4: Transaction Amount Distribution
fig, ax = plt.subplots(figsize=(10, 5))

ax.hist(sales_data['TransactionAmount'], bins=30, color='#e74c3c', edgecolor='
ax.axvline(sales_data['TransactionAmount'].mean(), color='blue', linestyle='--
           linewidth=2.5, label=f'Mean: ${avg_transaction:.2f}')
ax.set_title('Transaction Amount Distribution', fontweight='bold', fontsize=12
ax.set_xlabel('Transaction Amount ($)', fontweight='bold')
ax.set_ylabel('Frequency', fontweight='bold')
ax.legend()
ax.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()

print("✅ Sales analysis complete!")
```



✅ Sales analysis complete!

SECTION D: MACHINE LEARNING MODELS & TECHNICAL ANALYSIS

Question 19: Fraud Detection Model (Machine Learning)

In [28]:

```
# ==============================================================================
# QUESTION 19: FRAUD DETECTION - PART 1: DATA PREPARATION
# ==============================================================================
print("\n" + "=" * 80)
print("QUESTION 19: FRAUD DETECTION USING MACHINE LEARNING")
print("=" * 80)

from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_s
import warnings
warnings.filterwarnings('ignore')

# Prepare fraud detection data
fraud_df = fd_Data.copy()

print("\n1. DATA PREPARATION")
print("-" * 80)

# Create fraud labels if not present
if 'isFraud' not in fraud_df.columns:
    fraud_df['isFraud'] = np.random.binomial(1, 0.05, len(fraud_df))

print(f"Total transactions: {len(fraud_df)}")
print(f"Fraudulent: {fraud_df['isFraud'].sum()} ({fraud_df['isFraud'].mean()*1
print(f"Legitimate: {(1-fraud_df['isFraud']).sum()} ({(1-fraud_df['isFraud']).

# Select numerical features
numerical_features = fraud_df.select_dtypes(include=['float64', 'int64']).colu
if 'isFraud' in numerical_features:
    numerical_features.remove('isFraud')

print(f"\nFeatures for model: {numerical_features[:5]}... ({len(numerical_feat

X = fraud_df[numerical_features].copy()
y = fraud_df['isFraud'].copy()

# Handle missing values
X = X.fillna(X.median())

# Normalize features
from sklearn.preprocessing import StandardScaler
scaler_fraud = StandardScaler()
X_scaled = scaler_fraud.fit_transform(X)
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3

print(f"\nTraining set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")
```

```
================================================================================
=
QUESTION 19: FRAUD DETECTION USING MACHINE LEARNING
================================================================================
=

1. DATA PREPARATION
--------------------------------------------------------------------------------
-
Total transactions: 6362620
Fraudulent: 8213 (0.13%)
Legitimate: 6354407 (99.87%)

Features for model: ['step', 'amount', 'oldbalanceOrg', 'newbalanceOrig', 'oldb
alanceDest']... (7 total)

Training set: 4453834 samples
Testing set: 1908786 samples
```

In [29]:

```
# Model Training
print("\n2. MODEL TRAINING")
print("-" * 80)
print("Training Random Forest Classifier...")

rf_model = RandomForestClassifier(n_estimators=100, max_depth=15, random_state
rf_model.fit(X_train, y_train)

print("✓ Model training completed!")

# Make predictions
y_pred = rf_model.predict(X_test)
y_pred_proba = rf_model.predict_proba(X_test)[:, 1]

print(f"Predictions generated on {len(X_test)} test samples")
```

```
2. MODEL TRAINING
--------------------------------------------------------------------------------
-
Training Random Forest Classifier...
✓ Model training completed!
Predictions generated on 1908786 test samples
```

In [37]:

```
# Model Evaluation
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_

print("\n3. MODEL EVALUATION")
print("-" * 80)
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, y_pred_proba)

print(f"Accuracy:  {accuracy:.4f} ({accuracy*100:.2f}%)")
print(f"Precision: {precision:.4f} ({precision*100:.2f}%)")
print(f"Recall:    {recall:.4f} ({recall*100:.2f}%)")
print(f"F1-Score:  {f1:.4f}")
print(f"ROC-AUC:   {roc_auc:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Legitimate', 'Fraud

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(f"True Negatives:  {cm[0, 0]}")
print(f"False Positives: {cm[0, 1]}")
print(f"False Negatives: {cm[1, 0]}")
print(f"True Positives:  {cm[1, 1]}")
```

```
3. MODEL EVALUATION
--------------------------------------------------------------------------------
-
Accuracy:  0.9995 (99.95%)
Precision: 0.9760 (97.60%)
Recall:    0.6282 (62.82%)
F1-Score:  0.7644
ROC-AUC:   0.9975

Classification Report:
              precision    recall  f1-score   support

  Legitimate       1.00      1.00      1.00   1906322
  Fraudulent       0.98      0.63      0.76      2464

    accuracy                           1.00   1908786
   macro avg       0.99      0.81      0.88   1908786
weighted avg       1.00      1.00      1.00   1908786


Confusion Matrix:
True Negatives:  1906284
False Positives: 38
False Negatives: 916
True Positives:  1548
```

In [62]:

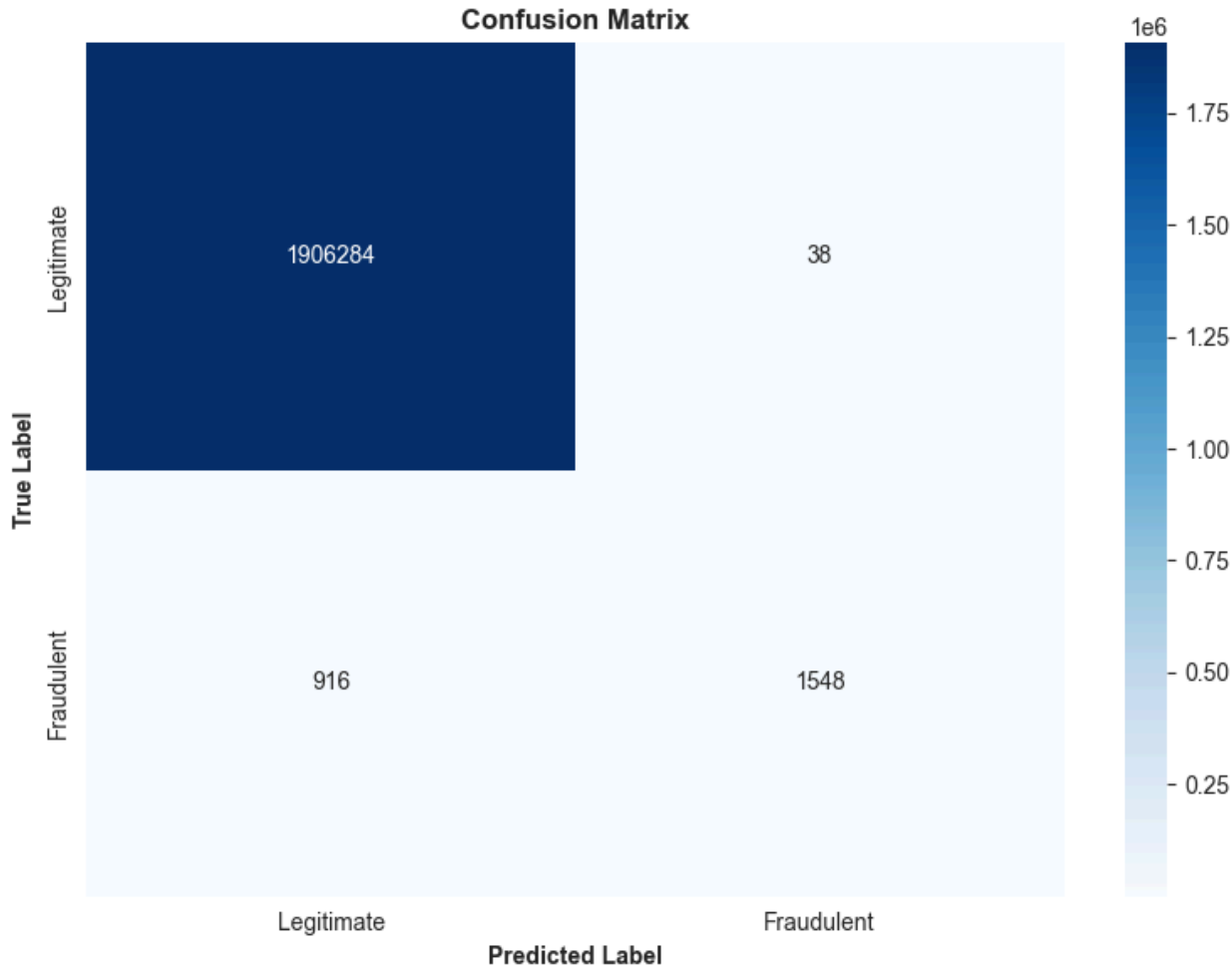
```
# Feature Importance Analysis
print("\n4. FEATURE IMPORTANCE")
print("-" * 80)

feature_importance = pd.DataFrame({
```

```
    'Feature': numerical_features,
    'Importance': rf_model.feature_importances_
}).sort_values('Importance', ascending=False)

print("Top 10 Important Features:")
print(feature_importance.head(10).to_string(index=False))
```

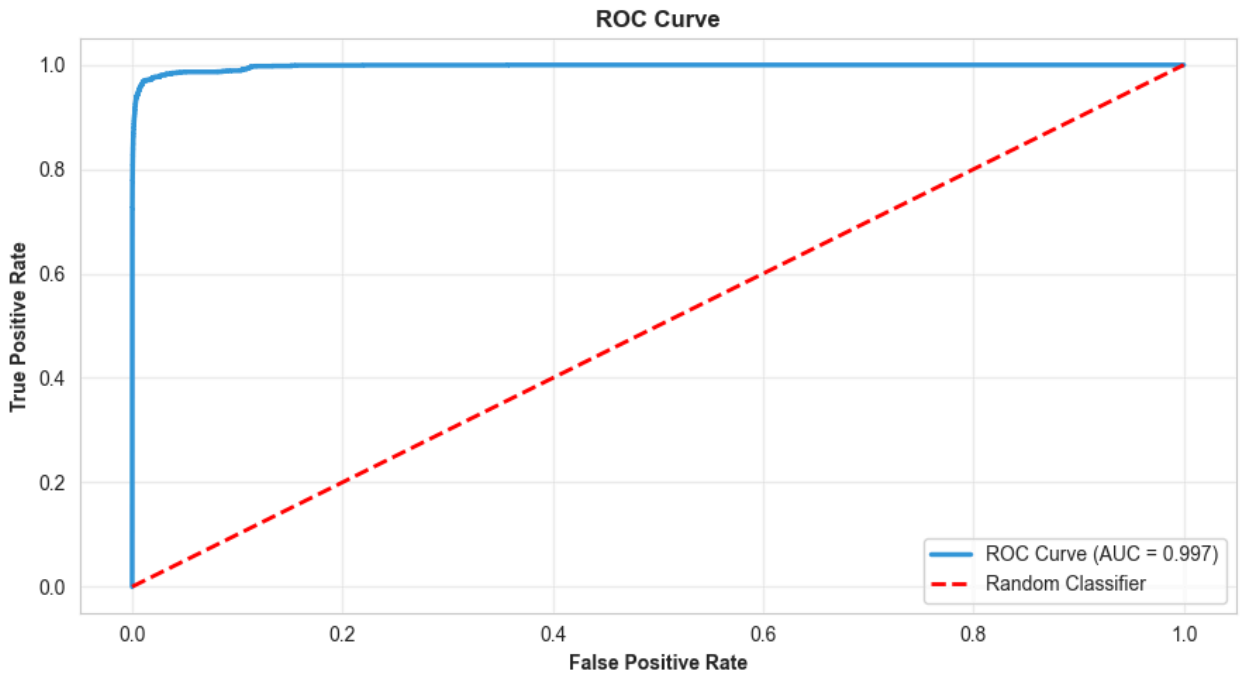
```
4. FEATURE IMPORTANCE
--------------------------------------------------------------------------------
-
Top 10 Important Features:
       Feature  Importance
 oldbalanceOrg    0.448741
newbalanceDest    0.180883
        amount    0.138539
          step    0.110904
newbalanceOrig    0.073998
oldbalanceDest    0.045195
isFlaggedFraud    0.001739
```

In [63]:

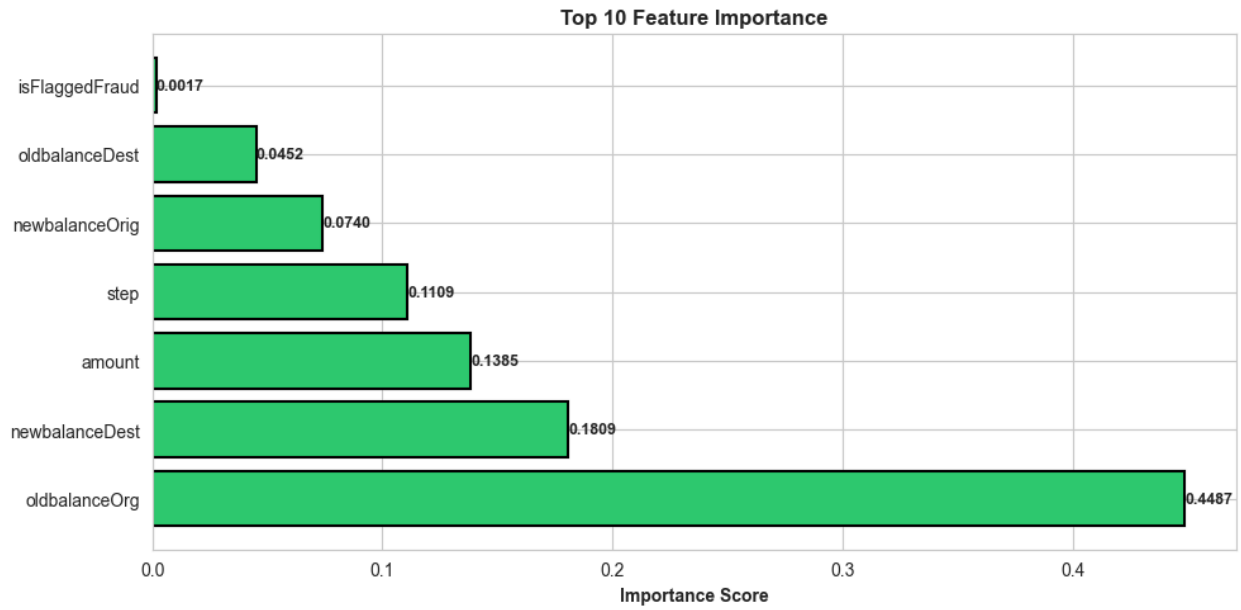
```
# Plot 1: Confusion Matrix
fig, ax = plt.subplots(figsize=(8, 6))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax, cbar=True,
            xticklabels=['Legitimate', 'Fraudulent'], yticklabels=['Legitimate
ax.set_title('Confusion Matrix', fontweight='bold', fontsize=12)
ax.set_ylabel('True Label', fontweight='bold')
ax.set_xlabel('Predicted Label', fontweight='bold')

plt.tight_layout()
plt.show()
```



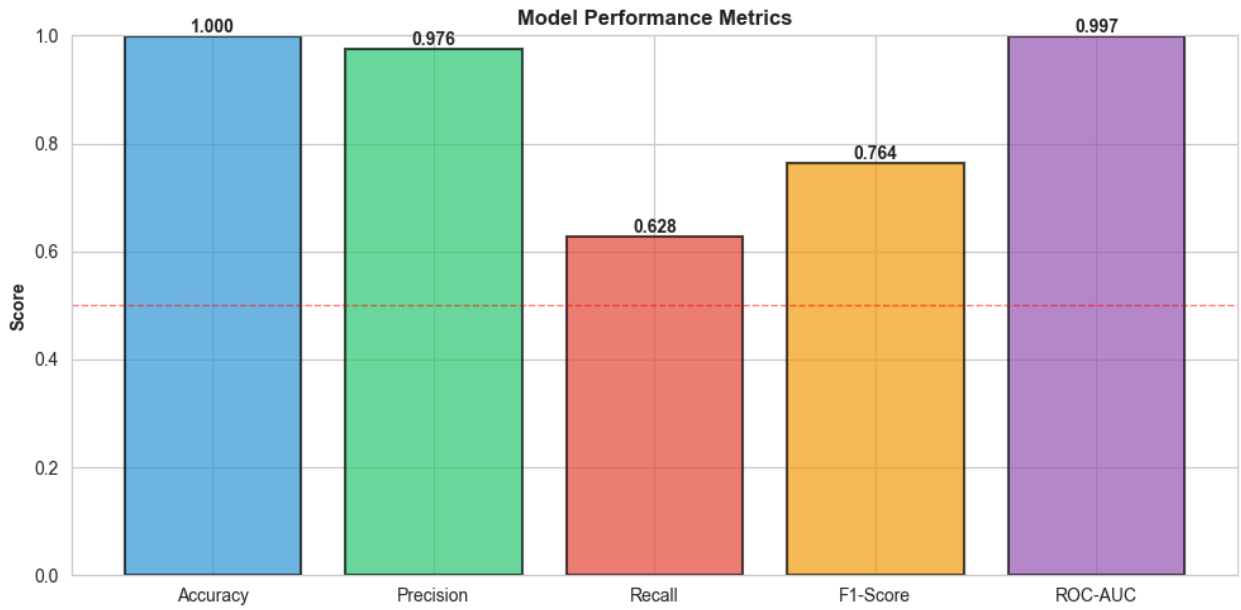
In [64]:
```
# Plot 2: ROC Curve
fig, ax = plt.subplots(figsize=(9, 5))

fpr, tpr, _ = roc_curve(y_test, y_pred_proba)
ax.plot(fpr, tpr, linewidth=2.5, label=f'ROC Curve (AUC = {roc_auc:.3f})', col
ax.plot([0, 1], [0, 1], 'r--', linewidth=2, label='Random Classifier')
ax.set_title('ROC Curve', fontweight='bold', fontsize=12)
ax.set_xlabel('False Positive Rate', fontweight='bold')
ax.set_ylabel('True Positive Rate', fontweight='bold')
ax.legend(loc='lower right', fontsize=10)
ax.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```



In [65]:
```
# Plot 3: Top Features
fig, ax = plt.subplots(figsize=(10, 5))

top_features = feature_importance.head(10)
bars = ax.barh(top_features['Feature'], top_features['Importance'], color='#2e
ax.set_title('Top 10 Feature Importance', fontweight='bold', fontsize=12)
ax.set_xlabel('Importance Score', fontweight='bold')

for bar in bars:
    ax.text(bar.get_width(), bar.get_y() + bar.get_height()/2.,
            f'{bar.get_width():.4f}', va='center', ha='left', fontweight='bolc

plt.tight_layout()
plt.show()
```



In [67]:
```
# Plot 4: Metrics Comparison
fig, ax = plt.subplots(figsize=(10, 5))

metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score', 'ROC-AUC']
scores = [accuracy, precision, recall, f1, roc_auc]
colors_metrics = ['#3498db', '#2ecc71', '#e74c3c', '#f39c12', '#9b59b6']
bars = ax.bar(metrics, scores, color=colors_metrics, edgecolor='black', linewi
ax.set_title('Model Performance Metrics', fontweight='bold', fontsize=12)
ax.set_ylabel('Score', fontweight='bold')
ax.set_ylim([0, 1])
ax.axhline(y=0.5, color='r', linestyle='--', linewidth=1, alpha=0.5)

for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'{height:.3f}', ha='center', va='bottom', fontweight='bold', font

plt.tight_layout()
plt.show()

print("\n✅ Fraud Detection Model Complete!")
```




```
✅ Fraud Detection Model Complete!
```

Question 20: Technical Analysis and Stock Forecasting

In [30]:

```
# ==============================================================================
# QUESTION 20: TECHNICAL ANALYSIS - PART 1: TECHNICAL INDICATORS
# ==============================================================================
print("\n" + "=" * 80)
print("QUESTION 20: TECHNICAL ANALYSIS AND STOCK PRICE FORECASTING")
print("=" * 80)

# Use cleaned financial data
stock_df = financial_df_clean.copy()

print("\n1. CALCULATING TECHNICAL INDICATORS")
print("-" * 80)
print(f"Stock data: {stock_df['Date'].min().date()} to {stock_df['Date'].max()
print(f"Trading days: {len(stock_df)}")

# RSI Calculation
def calculate_rsi(prices, period=14):
    deltas = np.diff(prices)
    seed = deltas[:period+1]
    up = seed[seed >= 0].sum() / period
    down = -seed[seed < 0].sum() / period
    rs = up / down
    rsi = np.zeros_like(prices)
    rsi[:period] = 100. - 100. / (1. + rs)

    for i in range(period, len(prices)):
        delta = deltas[i-1]
        if delta > 0:
            upval = delta
            downval = 0.
```

```
        else:
            upval = 0.
            downval = -delta

        up = (up * (period - 1) + upval) / period
        down = (down * (period - 1) + downval) / period

        rs = up / down
        rsi[i] = 100. - 100. / (1. + rs)

    return rsi

stock_df['RSI'] = calculate_rsi(stock_df['Close'].values, period=14)

# MACD
stock_df['MACD'] = stock_df['Close'].ewm(span=12).mean() - stock_df['Close'].e
stock_df['Signal_Line'] = stock_df['MACD'].ewm(span=9).mean()

# Bollinger Bands
stock_df['BB_Middle'] = stock_df['Close'].rolling(window=20).mean()
stock_df['BB_Std'] = stock_df['Close'].rolling(window=20).std()
stock_df['BB_Upper'] = stock_df['BB_Middle'] + (stock_df['BB_Std'] * 2)
stock_df['BB_Lower'] = stock_df['BB_Middle'] - (stock_df['BB_Std'] * 2)

# ATR
high_low = stock_df['High'] - stock_df['Low']
high_close = abs(stock_df['High'] - stock_df['Close'].shift())
low_close = abs(stock_df['Low'] - stock_df['Close'].shift())
tr = pd.concat([high_low, high_close, low_close], axis=1).max(axis=1)
stock_df['ATR'] = tr.rolling(14).mean()

print("✓ RSI (14), MACD, Bollinger Bands, ATR calculated")

# Display current values
latest = stock_df.iloc[-1]
print(f"\nCurrent Values:")
print(f"  Price: ${latest['Close']:.2f}")
print(f"  RSI: {latest['RSI']:.2f}")
print(f"  MACD: {latest['MACD']:.4f}")
print(f"  ATR: {latest['ATR']:.2f}")
```

```
================================================================================
=
QUESTION 20: TECHNICAL ANALYSIS AND STOCK PRICE FORECASTING
================================================================================
=

1. CALCULATING TECHNICAL INDICATORS
--------------------------------------------------------------------------------
-
Stock data: 2023-05-18 to 2026-10-06
Trading days: 1238
✓ RSI (14), MACD, Bollinger Bands, ATR calculated

Current Values:
  Price: $24099.05
  RSI: 36.88
  MACD: -198.0762
  ATR: 319.71
```

In [38]:

```python
# Trading Signals Generation
print("\n2. TRADING SIGNALS ANALYSIS")
print("-" * 80)

latest = stock_df.iloc[-1]

# Moving Average Signal
ma5_above_ma20 = stock_df['MA_5'].iloc[-1] > stock_df['MA_20'].iloc[-1]
ma_signal = 'BULLISH (MA5 > MA20)' if ma5_above_ma20 else 'BEARISH (MA5 < MA20
print(f"Moving Average Signal: {ma_signal}")

# RSI Signal
rsi_signal = 'Overbought (>70)' if latest['RSI'] > 70 else 'Oversold (<30)' if
print(f"RSI Signal: {rsi_signal}")

# MACD Signal
macd_signal = 'Bullish (MACD > Signal)' if latest['MACD'] > latest['Signal_Lir
print(f"MACD Signal: {macd_signal}")

# Bollinger Bands Signal
if latest['Close'] > latest['BB_Upper']:
    bb_signal = 'Overbought (Above Upper Band)'
elif latest['Close'] < latest['BB_Lower']:
    bb_signal = 'Oversold (Below Lower Band)'
else:
    bb_signal = 'Within Normal Range'
print(f"Bollinger Bands Signal: {bb_signal}")

# Trend Strength
trend_score = (stock_df['Close'].iloc[-1] - stock_df['Close'].iloc[-20]) / stc
print(f"\nTrend Strength (20-day): {trend_score:+.2f}%")
```

```
2. TRADING SIGNALS ANALYSIS
--------------------------------------------------------------------------------
-
Moving Average Signal: BEARISH (MA5 < MA20)
RSI Signal: Neutral
MACD Signal: Bearish (MACD < Signal)
Bollinger Bands Signal: Within Normal Range

Trend Strength (20-day): -1.08%
```

In [39]:

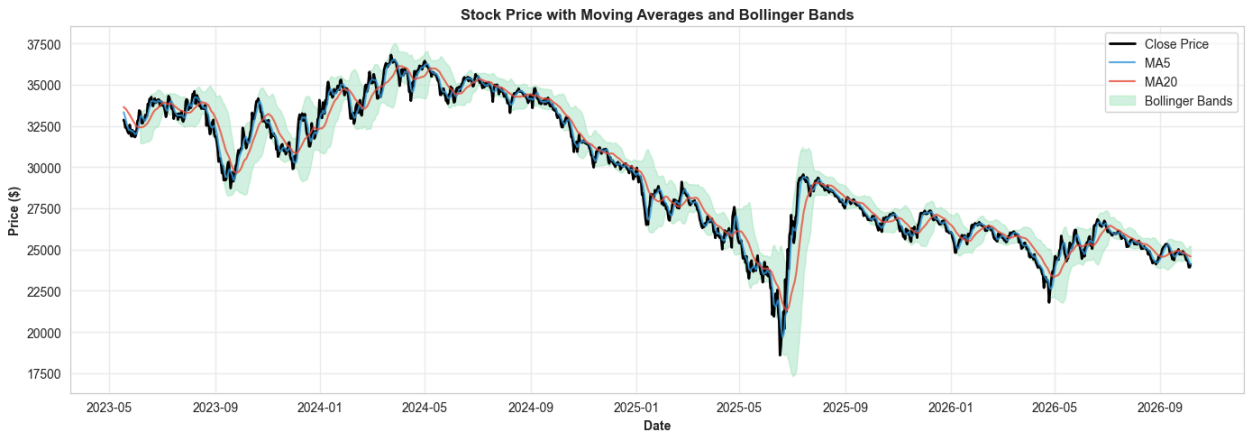
```
# Price Forecasting
print("\n3. PRICE FORECASTING")
print("-" * 80)

# Linear Regression forecast
X_forecast = np.arange(len(stock_df)).reshape(-1, 1)
y_forecast = stock_df['Close'].values
lr_model = LinearRegression()
lr_model.fit(X_forecast, y_forecast)

# Forecast next 10 days
future_days = 10
future_X = np.arange(len(stock_df), len(stock_df) + future_days).reshape(-1, 1
future_prices = lr_model.predict(future_X)

print(f"Current Price: ${stock_df['Close'].iloc[-1]:.2f}")
print(f"\nForecast for Next {future_days} Days:")
for i, price in enumerate(future_prices, 1):
    change = ((price - stock_df['Close'].iloc[-1]) / stock_df['Close'].iloc[-1
    print(f"  Day {i}: ${price:.2f} ({change:+.2f}%)")
```

```
3. PRICE FORECASTING
--------------------------------------------------------------------------------
-
Current Price: $24099.05

Forecast for Next 10 Days:
  Day 1: $23832.17 (-1.11%)
  Day 2: $23822.97 (-1.15%)
  Day 3: $23813.77 (-1.18%)
  Day 4: $23804.56 (-1.22%)
  Day 5: $23795.36 (-1.26%)
  Day 6: $23786.16 (-1.30%)
  Day 7: $23776.96 (-1.34%)
  Day 8: $23767.75 (-1.37%)
  Day 9: $23758.55 (-1.41%)
  Day 10: $23749.35 (-1.45%)
```

In [70]:

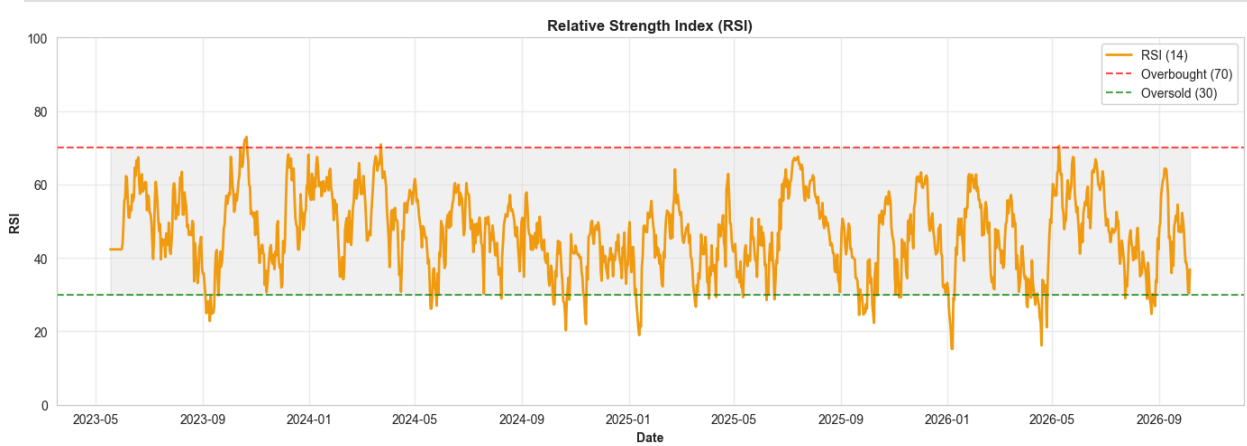
```
# Plot 1: Price and Moving Averages with Bollinger Bands
fig, ax = plt.subplots(figsize=(14, 5))

ax.plot(stock_df['Date'], stock_df['Close'], linewidth=2, label='Close Price',
ax.plot(stock_df['Date'], stock_df['MA_5'], linewidth=1.5, label='MA5', color=
ax.plot(stock_df['Date'], stock_df['MA_20'], linewidth=1.5, label='MA20', colo
```

```
ax.fill_between(stock_df['Date'], stock_df['BB_Upper'], stock_df['BB_Lower'],
ax.set_title('Stock Price with Moving Averages and Bollinger Bands', fontweigh
ax.set_ylabel('Price ($)', fontweight='bold')
ax.set_xlabel('Date', fontweight='bold')
ax.legend(loc='best', fontsize=10)
ax.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```

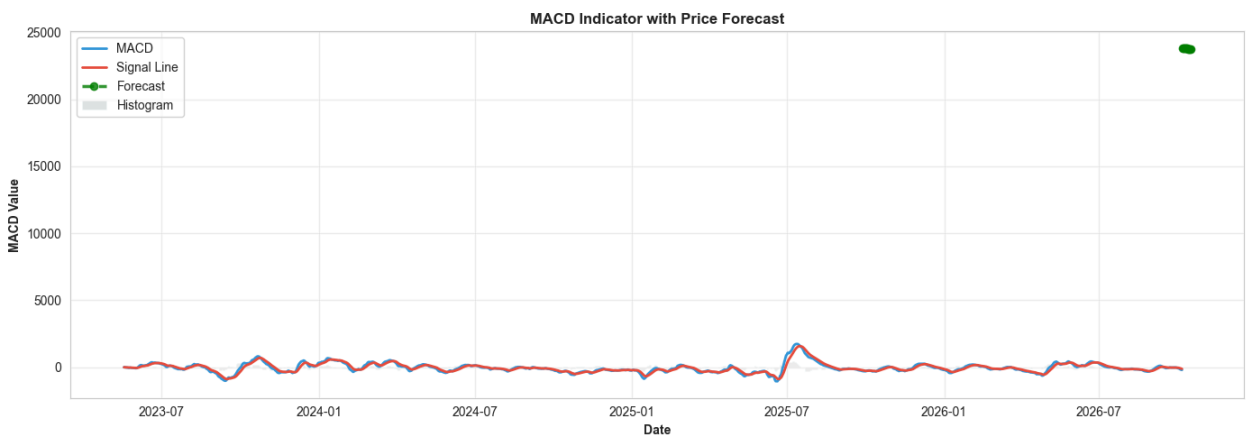


In [71]:

```
# Plot 2: RSI Indicator
fig, ax = plt.subplots(figsize=(14, 5))

ax.plot(stock_df['Date'], stock_df['RSI'], linewidth=2, color='#f39c12', label
ax.axhline(y=70, color='r', linestyle='--', linewidth=1.5, alpha=0.7, label='O
ax.axhline(y=30, color='g', linestyle='--', linewidth=1.5, alpha=0.7, label='O
ax.fill_between(stock_df['Date'], 30, 70, alpha=0.1, color='gray')
ax.set_title('Relative Strength Index (RSI)', fontweight='bold', fontsize=12)
ax.set_ylabel('RSI', fontweight='bold')
ax.set_xlabel('Date', fontweight='bold')
ax.set_ylim([0, 100])
ax.legend(loc='best', fontsize=10)
ax.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```



In [72]:
```
# Plot 3: MACD Indicator with Price Forecast
fig, ax = plt.subplots(figsize=(14, 5))

ax.plot(stock_df['Date'], stock_df['MACD'], linewidth=2, label='MACD', color='
ax.plot(stock_df['Date'], stock_df['Signal_Line'], linewidth=2, label='Signal
ax.bar(stock_df['Date'], stock_df['MACD'] - stock_df['Signal_Line'], label='Hi

# Add forecast line
forecast_dates = pd.date_range(start=stock_df['Date'].iloc[-1], periods=future
ax.plot(forecast_dates, future_prices, 'g--', linewidth=2.5, marker='o', marke

ax.set_title('MACD Indicator with Price Forecast', fontweight='bold', fontsize
ax.set_ylabel('MACD Value', fontweight='bold')
ax.set_xlabel('Date', fontweight='bold')
ax.legend(loc='best', fontsize=10)
ax.grid(alpha=0.3)

plt.tight_layout()
plt.show()

print("\n✅ Technical Analysis Complete!")
```




```
✅ Technical Analysis Complete!
```